Quantum Error-Correcting Codes from Galois Hulls of Cyclic Codes over Noncommutative Matrix Rings

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Abstract

We construct q-ary quantum codes from the Galois hulls of cyclic codes over the noncommutative matrix ring $R = M_2(A_t)$, $A_t = \mathbb{F}_q[u]/\langle u^t\rangle$; codes are right ideals of $R_n = R[X]/\langle X^n - 1\rangle$. Duality is driven by the anti-automorphism $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$, σ_s the coefficient Frobenius. We prove the duality and hull theory for arbitrary right ideals, with no principality hypothesis, and refute the conjecture that Galois hulls are principal: hulls with minimal generator number $\mu_B = 3$ exist, and $\mu_B \geq 3$ forces $t \geq 2$ and $p \mid n$. The coefficient-expansion map to $\mathbb{F}_q^{4tn}$ transports the Galois pairing to an explicit $\widehat{\sigma}_s$-Hermitian form with Hankel Gram matrix $I_{4n} \otimes H_t$ — not the Euclidean form unless $t = 1$ — and preserves duality and hulls, including the non-principal ones. Every Galois hull is self-orthogonal for the transported form, with no reflexivity hypothesis. Three quantum routes of length $N = 4tn$ follow: a twisted stabilizer route, a Euclidean/CSS route, and an entanglement-assisted route $[[N, N-k-h_E, d; k-h_E]]_q$. We show that the distance is the minimum weight of the quotient $D^{\perp_E}/\operatorname{Hull}_E(D)$, and derive a coset Singleton bound $K + d \leq N + 1$ not implied by the entanglement-assisted Singleton bound. Exhaustive computation certifies 28 codes by three independent methods, including this framework's first distance-4 stabilizer codes, $[[24, 8, 4]]_9$ and $[[20, 4, 4]]_4$, and entanglement-assisted codes up to $[[20, 2, 10; 18]]_4$. No record parameters are claimed.

Keywords. cyclic codes over rings; matrix rings; Galois duality; Galois hulls; Gray map; quantum stabilizer codes; entanglement-assisted quantum codes.

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1 Introduction and research motivation

The construction of quantum error-correcting codes from classical codes over rings is by now a mature enterprise, and one of its most productive branches runs through the *hull* of a code: for a code C equipped with a pairing $\langle \cdot, \cdot \rangle$, the hull $C \cap C^{\perp}$ measures how far C is from being self-orthogonal, and it is precisely the hull dimension that determines the parameters of the quantum code one can extract. Euclidean hulls give CSS codes, Hermitian hulls give q-ary stabilizer codes, and — when the hull is small rather than large — the entanglement-assisted formalism converts *any* classical code into a quantum code at the price of shared entanglement.

Parallel to this, the Galois (or ℓ-Galois) hull has been introduced over finite fields [4, 9] and used to produce entanglement-assisted codes with good parameters [2]. The Galois parameter ℓ interpolates between the Euclidean and the Hermitian inner product, and because it can be tuned it gives a strictly larger supply of codes with a prescribed hull dimension.

What has been missing is the passage to a *non-commutative* alphabet. There is a well-developed theory of cyclic codes over matrix rings [1, 10, 3, 7], and there is a well-developed theory of quantum codes from Galois hulls over commutative rings, culminating in the recent constacyclic construction over a finite non-chain ring [13]. The two have not met, and they do not meet trivially: over a commutative ring the Galois twist is an automorphism applied to one argument of the inner product, whereas over a matrix ring the twist must be composed with the transpose to produce an *anti*-automorphism, and only then does the resulting pairing give a perfect duality on *right* modules. This paper builds that bridge and constructs quantum codes at the far end of it.

1.1 The chain of constructions

Throughout, $q = p^r$, and

$$A_t = \mathbb{F}_q[u]/\langle u^t \rangle, \qquad R = M_2(A_t), \qquad R_n = R[X]/\langle X^n - 1 \rangle, \qquad N = 4tn.$$

A *cyclic code of length n over R* is a right ideal of R_n; this is the right-module convention fixed for this alphabet in [1], and Lemma 3.4 below shows that it is not a cosmetic choice. Under the Morita identification $R_n \cong M_2(B)$ with $B = A_t[X]/\langle X^n - 1\rangle$ *commutative*, right ideals of R_n correspond to B-submodules of B^2.

The contribution of this paper is the following chain, each link of which is proved here:

$$\boxed{\text{noncommutative matrix ring}} \longrightarrow \boxed{\text{cyclic codes}} \longrightarrow \boxed{\text{Galois duality and hulls}}$$
$$\longrightarrow \boxed{\text{Gray map}} \longrightarrow \boxed{\text{quantum codes}}.$$

In more detail:

(1) **Galois duality on a non-commutative ring.** With $\sigma_s(a) = a^{p^s}$ the coefficient Frobenius (fixing u and X) and $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$, the pairing $\mathbf{b}_s(x, y) = \Lambda\big(\sum_i \theta_s(x_i)y_i\big)$, where $\Lambda = \lambda \circ \mathrm{tr}$ and λ extracts the coefficient of u^{t-1}, is non-degenerate on right R_n-modules, and $\dim C + \dim C^{\perp_s} = N$ for every code (Theorem 4.3). We give the dual explicitly for *arbitrary* finitely generated right ideals (Theorem 4.6), including the reversal operator, which is load-bearing exactly when $n \geq 3$.

(2) **Hulls, with no principality hypothesis.** We prove the structural properties of $\mathrm{Hull}_s(C) = C \cap C^{\perp_s}$ for arbitrary right ideals, and we *refute* the natural conjecture that a Galois hull is principal: hulls with minimal generator number $\mu_B = 3$ exist (Theorem 5.4). The phenomenon is confined: $\mu_B \geq 3$ forces $t \geq 2$ and $p \mid n$.

(3) **The exact induced form.** The coefficient expansion $\varphi_n : R^n \to \mathbb{F}_q^N$ is not an isometry in any interesting sense, but it *is* a form-isometry for exactly one form, and we identify that form: $\langle u, v\rangle_s^{\frown} = u^{[s]} J v^{\mathsf{T}}$ with $J = I_{4n} \otimes H_t$ and H_t the Hankel reversal matrix (Theorem 6.3). Duality and hulls are transported exactly, and the transport is valid on the non-principal hulls as well (Theorem 6.6).

(4) **Every Galois hull is self-orthogonal** for the transported form, by a one-sided argument that uses neither reflexivity nor principality (Theorem 6.10). This is the hinge of the whole construction: it is what makes the quantum step available in the general Galois regime rather than only in the reflexive one.

(5) **Quantum codes of length $N = 4tn$.** Three routes are available and all three are proved from scratch in symplectic language: a Galois-native twisted stabilizer route R1, a Euclidean/CSS route R2 and an entanglement-assisted route R3 giving $[[N, N - k - h_E, d; k - h_E]]_q$ (Theorem 7.9).

(6) **A coset-distance theorem and a new bound.** The distance is the minimum weight of the *quotient* code $D^{\perp_E} / \mathrm{Hull}_E(D)$, of dimension K and length N (Theorem 8.1); puncturing $N - d + 1$ coordinates is injective on that quotient, which gives the *coset Singleton bound* $K + d \leq N + 1$ (Theorem 8.3). The latter is neither the entanglement-assisted Singleton bound nor implied by it.

1.2 What the paper does and does not claim

The codes constructed here are new in *origin*, not in parameters. Exhaustive computation over the accessible family list ($q \in \{4, 8, 9\}$, $tn \leq 8$) certifies 28 codes by three independent methods, among them the first stabilizer distance 4 produced by this framework, $[[24, 8, 4]]_9$ and $[[20, 4, 4]]_4$, and entanglement-assisted codes up to $[[20, 2, 10; 18]]_4$. Every stabilizer code we construct whose parameters can be compared with [5] is at best equal to the best known distance — and then only with $d = 2$ — and every code with $d \geq 3$ is one or two below it. We

claim no record. What we do claim is that the chain above is proved in the non-commutative setting, that it produces quantum codes from hulls that are *not* principal, and that the two new bounds of item (6) hold.

1.3 The role of the matrix-ring formulation, and what Morita equivalence does and does not settle

Since X is central, R_n is canonically isomorphic to $M_2(B)$ with $B = A_t[X]/\langle X^n - 1\rangle$ *commutative*, and under the Morita correspondence right ideals of R_n correspond to B-submodules of B^2; the hull of a code is then the ideal corresponding to the intersection of the two associated submodules (Proposition 5.3). It is therefore fair to ask what the matrix-ring formulation contributes. Three points.

First, Morita equivalence is an equivalence of module categories, not of rings with involution, and it does not carry the duality. The duality used throughout is defined by the anti-automorphism $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$ of R, and it is the transpose that makes the dual of a *right* ideal a right ideal with the correct size (Lemma 3.4 and Theorem 4.3). Over a commutative coefficient ring the twist degenerates to σ_s and there is nothing to prove. The pairing of Definition 4.1, the distinction between the general Galois and the reflexive regime (Definition 4.9), and the self-orthogonality of hulls without reflexivity (Theorem 6.10) are all statements about R and θ_s made *before* any Morita translation.

Second, what Morita does give is exactly what we use it for: it converts the structure theory of hulls into a computation with submodules of B^2. That is how the principality criterion of Proposition 5.3 and the confinement of Theorem 5.4 are proved, and how the witnesses are constructed. *We do not claim novelty from the non-commutativity of $M_2(A_t)$ alone.* The novelty claimed is in the duality that the non-commutative alphabet forces, and in the hull phenomena that the ($t \geq 2$ and $p \mid n$) criterion isolates.

Third, after the translation the residual problem is a problem about $B = \mathbb{F}_q[u, X]/\langle u^t, X^n - 1\rangle$. Its Chinese-remainder decomposition is governed by the factorisation of $X^n - 1$ over $\mathbb{F}_q$, and the condition $p \mid n$ is precisely the repeated-root case. This is where the finite-field content of the paper lives, and it is why the framework is reported here rather than only in the language of matrices over a chain ring.

1.4 Contributions, in order of emphasis

The contributions, in the order in which we ask them to be read, are:

1. **An algebraic Galois-duality and hull framework over a non-commutative alphabet.** The Galois twist must be the anti-automorphism θ_s, duality and hull theory are developed for arbitrary right ideals with no principality hypothesis, and self-orthogonality holds with no reflexivity hypothesis (Section 4, Section 5).

2. **Non-principal hull structure, and a counterexample.** Galois hulls need not be principal; $\mu_B \geq 3$ forces $t \geq 2$ and $p \mid n$; and no closed formula of gcd type exists (Theorem 5.4, Theorem 5.8).

3. **A duality-preserving expansion to $\mathbb{F}_q^N$.** The transported form is identified exactly, with Hankel Gram matrix $I_{4n} \otimes H_t$, and duality and hulls are preserved, non-principal ones included (Section 6).

4. **Quantum stabilizer and entanglement-assisted constructions.** Three routes of length $N = 4tn$, with the characteristic-two obstruction of Theorem 6.13 delimiting the Euclidean route (Section 7).

5. **Exact distance certificates and two bounds.** The coset-distance theorem and the coset Singleton bound $K + d \leq N + 1$, and the certification of 28 codes by three independent methods (Section 8).

The enumerators and dimension-distribution statistics produced by the scan are **supporting material**. They are reported where they support a construction and are not presented as principal results.

1.5 On the parameters obtained

Every stabilizer code constructed here is at best *equal* to the best known distance, and then only at $d = 2$; every stabilizer code with $d \geq 3$ is one or two below the best known value (Section 9). The entanglement-assisted codes carry the largest distances obtained in this framework, and they cannot be benchmarked at all, because no entanglement-assisted table is available; the codes over $\mathbb{F}_8$ are likewise unbenchmarked. **No record parameter is claimed.** What is offered is an exact algebraic *origin* for these codes: non-commutative Galois hulls, transported to $\mathbb{F}_q$ with duality and hulls preserved, with non-principal hulls admitted, and with two new bounds constraining the parameters that result. The value of the construction is that every parameter reported is derived and certified rather than merely asserted, and that the algebraic hypotheses behind each tuple are visible.

1.6 Organisation

Section 2 reviews the literature and states the research gap. Section 3 fixes the rings, the right-module convention and the anti-automorphism, and proves the failure lemma that motivates the convention. Section 4 develops cyclic codes and Galois duality. Section 5 is the algebraic heart: hull structure, the non-principal counterexample and its confinement. Section 6 transports everything to $\mathbb{F}_q^N$ and identifies the induced form. Section 7 builds the quantum codes. Section 8 proves the coset-distance theorem and the coset Singleton bound, describes the certification methodology and reports the parameter analysis. Section 9 compares with the literature and states the novelty precisely. Section 10 collects open problems. Appendix A contains the full parameter tables and Appendix B the reproducibility package.

2 Literature review and the research gap

2.1 Cyclic codes over matrix rings

The use of a matrix ring as a code alphabet goes back to Bachoc's construction of modular lattices, and the coding-theoretic study of cyclic codes over $M_2(\mathbb{F}_2)$ was initiated by Alahmadi, Sboui, Solé and Yemen [1], who fixed the convention that cyclic codes are *right* ideals and used the ring-valued Euclidean dot product. Luo and Parampalli [10] extended the structure theorem to $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$, i.e. to $t = 2$ in our notation, and Dinh, Gaur, Kumar, Singh and Singh [3] to $M_2(\mathbb{F}_2[u]/\langle u^k \rangle)$ for all k — the same ring family as the present paper — obtaining the structure via an isometry onto $\mathbb{F}_4[u, v]/\langle u^k, v^2 \rangle$. Most recently Joshi, Patel and Prakash [7] treated $M_4(\mathbb{F}_2[u]/\langle u^k \rangle)$ with Euclidean and Hermitian duals and both the Bachoc and the Gray map into $\mathbb{F}_{16}$. A separate, genuinely different line uses skew polynomial rings and cyclic algebras over $M_2(\mathbb{F}_2)$ [11].

What *none* of these papers contains is a Galois duality, a hull, or a quantum construction. Their duals are the Euclidean and the Hermitian dual, and their output is a classical code over a finite field, compared with the best known classical codes.

2.2 Duality over finite rings

The authority for duality over non-commutative finite rings is Wood [12]: the extension of the MacWilliams equivalence theorem to finite rings characterises Frobenius rings, and duality for modules over a finite Frobenius ring is the correct general framework. We import the statement that $R = M_m(A_t)$ is Frobenius and prove directly, in Lemma 3.3, that $\Lambda = \lambda \circ \mathrm{tr}$ is a Frobenius functional on it.

For the weight side, Greferath and Schmidt [6] construct Gray isometries for finite *chain* rings. Their theorem does not apply to $M_2(A_t)$, which is not a chain ring for $t \geq 1$ and $m = 2$; we use it only to delimit what our own map can achieve (Remark 6.9).

2.3 Galois duality and Galois hulls

Over a finite field, Fan and Zhang [4] introduced the ℓ-Galois inner product $\langle a, b\rangle_\ell = \sum_i a_i b_i^{p^\ell}$ and studied ℓ-Galois self-dual constacyclic codes; Liu and Pan [9] introduced the ℓ-Galois hull $\mathrm{Hull}_\ell(C) = C \cap C^{\perp_\ell}$ and gave its dimension formulas. Cao [2] constructed MDS codes with Galois hulls of arbitrary dimensions and the corresponding entanglement-assisted quantum codes. All of this takes place over a *commutative* base.

The closest prior work to the present paper is Zhang, Kong and Zheng [13], who study constacyclic codes over a finite non-chain ring R (commutative, with a decomposition $C = e_1C_1 \oplus \cdots \oplus e_tC_t$), determine the dimensions of the ℓ-Galois hulls of their Gray images, and construct EAQECCs and QECCs by a construction X for the Galois inner product, obtaining some QMDS codes and some QECCs better than the best known ones.

2.4 The research gap

The gap is the conjunction of three properties, none of which is present in the literature:

(1) the alphabet is **non-commutative** (a matrix ring over a finite chain ring), so that the Galois twist must be an *anti*-automorphism and the left/right distinction is substantive;

(2) the duality is **Galois** (a Frobenius parameter s is present, interpolating between the Euclidean and Hermitian cases in a way that a commutative alphabet cannot reproduce);

(3) the outcome is **quantum**: stabilizer and entanglement-assisted codes, obtained from the hull through a Gray map whose induced form is identified exactly.

Property (1) alone is the matrix-ring literature; (2) alone is the Galois-hull literature; (2)+(3) over a commutative non-chain ring is [13]. The conjunction (1)+(2)+(3) is what this paper supplies.

Two consequences of the non-commutativity are worth signalling here, because they have no commutative analogue and they shape the whole paper.

- The pairing must use $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$, not σ_s. Over a commutative ring the transpose is invisible and the ℓ-Galois form is simply Frobenius-in-one-argument; here the transpose is what makes θ_s an anti-automorphism, and the anti-automorphism is what makes the dual of a *right* module again a right module (Theorem 4.3). The plain product $\sum_i x_i y_i$, which is the Euclidean dot product used in [1], has a perfect dual on left modules but *not* on right modules (Lemma 3.4).

- The hull need not be principal, so no generator-polynomial formula for the hull can exist, and every downstream statement must be proved without a principality hypothesis. This is not a technical nuisance: the counterexamples are exactly the families that the quantum constructions later use (Theorem 5.4 and Section 5.3).

2.5 Quantum codes from classical codes

The stabilizer formalism for q-ary quantum codes is due to [8] in the non-binary case. We do not import the parameter formulas: Section 7 re-derives the stabilizer lemma and all three routes from the symplectic commutation relation, so that every hypothesis (in particular the requirement that commutation be tested on an $\mathbb{F}_p$-basis, Theorem 7.3) is visible where it is used.

3 Preliminaries and the noncommutative matrix-ring framework

3.1 Standing hypotheses and notation

Throughout, p is a prime, $q = p^r$, and $t, n \geq 1$ are integers. We set

$$A_t = \mathbb{F}_q[u]/\langle u^t \rangle, \qquad R = M_2(A_t), \qquad B = A_t[X]/\langle X^n - 1 \rangle, \qquad R_n = R[X]/\langle X^n - 1 \rangle, \quad (1)$$

and write $N = 4tn$. Elements of A_t are written $\sum_{a=0}^{t-1} c_a u^a$. All dimensions and all cardinalities of codes are over $\mathbb{F}_q$ unless a different field is named explicitly; this is maintained without exception, and the only place where $\mathbb{F}_p$ enters is Theorem 7.3.

Definition 3.1 (cyclic code)**.** A *cyclic code of length n over R* is a right ideal of R_n. We identify R_n with R^n by $x = \sum_e x_e X^e \leftrightarrow (x_0, \ldots, x_{n-1})$ and write $C \leq R_n$ for a right R_n-submodule.

The choice of right ideals is forced, not conventional: see Lemma 3.4. The ring B is commutative, being a quotient of the commutative ring $A_t[X]$, and there is a canonical isomorphism

$$R_n \;\cong\; M_2(B) \quad (2)$$

because X is central, so that a polynomial with matrix coefficients may be read as a matrix of polynomials. Under (2), right ideals of R_n correspond to B-submodules of B^2; this is the Morita correspondence we use repeatedly and we do not reprove it.

Lemma 3.2 (basic structure)**.** (a) *A_t is a finite chain ring: its ideals are exactly $u^i A_t$, $0 \leq i \leq t$; $|A_t| = q^t$; its maximal ideal is uA_t and its residue field is $\mathbb{F}_q$.*

(b) *R is a finite Frobenius ring with $|R| = q^{4t}$, $\dim_{\mathbb{F}_q} R = 4t$, $Z(R) = A_t \cdot I$, $\mathrm{Jac}(R) = M_2(uA_t)$, and exactly $t+1$ two-sided ideals, namely $u^i R$.*

(c) *$\dim_{\mathbb{F}_q} R^n = N = 4tn$.*

Proof. (a) Every element of A_t has a unique form $c_i u^i +$ (higher terms) with $c_i \neq 0$, and then the element is u^i times a unit; hence every ideal is of the stated form and they form a chain. (b) The stated descriptions of the centre and the Jacobson radical are immediate from the matrix-unit computation; the two-sided ideals of $M_2(S)$ are $M_2(I)$ for ideals $I \trianglelefteq S$, which gives $t+1$ of them. The Frobenius property is imported [12]; a direct Frobenius functional is exhibited in Lemma 3.3 below. (c) is immediate from (b), since $\dim_{\mathbb{F}_q} A_t = t$ and $\dim_{\mathbb{F}_q} R = 4 \dim_{\mathbb{F}_q} A_t$. □

3.2 The Frobenius functional

Let $\lambda : A_t \to \mathbb{F}_q$ be the coefficient of u^{t-1}, and let tr denote the matrix trace. Put

$$\Lambda = \lambda \circ \mathrm{tr} : R \longrightarrow \mathbb{F}_q, \qquad \Lambda(X) = \lambda(X_{11} + X_{22}). \quad (3)$$

Lemma 3.3. *Λ is symmetric ($\Lambda(XY) = \Lambda(YX)$), associative (the kernel contains no nonzero left or right ideal), $\mathbb{F}_q$-linear, and non-degenerate: the bilinear form $(X, Y) \mapsto \Lambda(XY)$ on R is non-degenerate.*

Proof. Symmetry: $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ for matrices over any commutative ring, and λ is $\mathbb{F}_q$-linear, so $\Lambda(XY) = \Lambda(YX)$.

Non-degeneracy: let $X = (X_{ij}) \neq 0$ and choose (k, l) with $X_{kl} \neq 0$; write $X_{kl} = \sum_a c_a u^a$ and let a_0 be the least index with $c_{a_0} \neq 0$. Take Y to be the matrix whose (l, k) entry is u^{t-1-a_0} and whose other entries are 0. Then XY has trace $\sum_{i,j} X_{ij} Y_{ji} = X_{kl} u^{t-1-a_0}$, whose u^{t-1}-coefficient is $c_{a_0} \neq 0$. Hence $\Lambda(XY) \neq 0$, and the form is non-degenerate.

Associativity (no nonzero one-sided ideal lies in $\ker \Lambda$): if $X \neq 0$, the Y constructed above already gives $\Lambda(XY) \neq 0$ with Y arbitrary in R, so the right annihilator of Λ in R is zero; by symmetry the same holds on the left. □

Together with the import that R is Frobenius, Lemma 3.3 says that (R, Λ) is a Frobenius algebra over $\mathbb{F}_q$, which is the setting we need.

3.3 The right-module convention and the failure lemma

Lemma 3.4 (failure lemma)**.** *Let S be a finite Frobenius ring and let C be a **left** ideal. Then the dual of C with respect to the plain product $\sum_i x_i y_i$ satisfies $|C| \cdot |C^\perp| = |S|^n$. For a **right** ideal this fails in general. Concretely, for $S = R = M_2(\mathbb{F}_2)$, $C = uR$ (with $t = 2$, $q = 2$, $n = 1$, so that u denotes a nilpotent generator of A_2 embedded diagonally) one has $|C| \cdot |C^\perp| = 4 \neq 16 = |R|$.*

Proof. For a left ideal C, the map $\psi_C : C \to \mathrm{Hom}_{\mathbb{F}_q}(S/C^\perp, \mathbb{F}_q)$, $c \mapsto (y \mapsto \Lambda(cy))$, is an $\mathbb{F}_q$-linear isomorphism by the non-degeneracy of Λ, and the standard dimension count gives the size formula.

For right ideals the argument breaks: ψ_C is no longer $\mathbb{F}_q$-balanced in the way required, because the plain product is not compatible with the right action. The stated counterexample is a direct computation: with $R = M_2(\mathbb{F}_2)$ and $C = uR$ one has $|C| = 4$ while the plain-product dual $C^\perp = \{y : \mathrm{tr}(xy) = 0 \ \forall x \in C\}$ has size 1, giving $|C||C^\perp| = 4 \neq 16$. □

Remark 3.5. Lemma 3.4 is the reason the pairing of Definition 4.1 below carries the transpose: only then does the dual of a right module come out as a right module with the correct size (Theorem 4.3). This is not a matter of notation.

3.4 Automorphisms and the anti-automorphism

Definition 3.6. For $s \geq 0$ let $\sigma_s : \mathbb{F}_q \to \mathbb{F}_q$, $a \mapsto a^{p^s}$. Extend σ_s to A_t, to R and to R_n coefficient-wise, fixing u and X. Define

$$\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}, \tag{4}$$

the composite of the coefficient-wise σ_s with the transpose.

Lemma 3.7. (a) *$\sigma_s \in \mathrm{Aut}(A_t)$ and its entrywise extension lies in $\mathrm{Aut}(R)$; σ_s is nontrivial iff $r \nmid s$, and its order is $r/\gcd(r, s)$.*

(b) *θ_s is an **anti**-automorphism of $R[X]$ and of R_n: $\theta_s(ab) = \theta_s(b)\theta_s(a)$.*

(c) *$\theta_s^2 = \sigma_{2s}$; more generally $\theta_s^{2k} = \sigma_{2ks}$ and $\theta_s^{2k-1} = \theta_{(2k-1)s}$. In particular θ_s is an involution iff $r \mid 2s$.*

(d) *$\theta_s(X^n - 1) = X^n - 1$, so θ_s is well defined on R_n; it is additive, $\mathbb{F}_p$-linear and σ_s-semilinear over $\mathbb{F}_q$ (not $\mathbb{F}_q$-linear unless $r \mid s$).*

Proof. (a) $a \mapsto a^{p^s}$ is a field automorphism of $\mathbb{F}_q$ of order $r/\gcd(r,s)$, and it fixes the coefficients of the relation $u^t = 0$, so it extends to A_t by fixing u; entrywise extension preserves matrix multiplication.

(b) For matrices over any ring, $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$, and σ_s is a ring automorphism, hence $\theta_s(ab) = \sigma_s((ab)^{\mathsf{T}}) = \sigma_s(b^{\mathsf{T}})\sigma_s(a^{\mathsf{T}}) = \theta_s(b)\theta_s(a)$. The same computation works in $R[X]$; note that there the index of X is swapped as well, which is why the statement is recorded separately for $R[X]$ and for R_n.

(c) $\theta_s^2(X) = \sigma_s(\sigma_s(X^{\mathsf{T}})^{\mathsf{T}}) = \sigma_s^2(X) = \sigma_{2s}(X)$; the general powers follow by induction.

(d) σ_s fixes X and the coefficients of $X^n - 1$ are $0, \pm 1 \in \mathbb{F}_p$, which σ_s fixes. Semilinearity is immediate from σ_s being a field automorphism of order > 1 in general. □

Remark 3.8. We use only the coefficient-Frobenius subgroup of $\mathrm{Aut}(A_t)$: an automorphism of A_t may move u, and we never use those. Everything in this paper generalises verbatim to $\tau \in \mathrm{Aut}(A_t)$ with $\theta_\tau = \tau \circ (\cdot)^{\mathsf{T}}$ and $\theta_\tau^2 = \tau^2$; we keep σ_s for concreteness.

3.5 The CRT decomposition of B

Let $X^n - 1 = \prod_j g_j^{e_j}$ be the factorisation into pairwise coprime monic irreducibles over $\mathbb{F}_q$. By the Chinese remainder theorem,

$$B = A_t[X]/\langle X^n - 1\rangle \;\cong\; \prod_j B_j, \qquad B_j = A_t[X]/\langle g_j^{e_j}\rangle, \tag{5}$$

and, writing $d_j = \deg g_j$, one has $B_j \cong \mathbb{F}_{q^{d_j}}[u,w]/\langle u^t, w^{e_j}\rangle$. The isomorphism (5) is realised explicitly as an $\mathbb{F}_q$-linear ring isomorphism Ψ (with matrix and inverse), so that it composes with the Gray map, the Morita identification and the quantum routes.

Lemma 3.9 (component generation)**.** *Let $W \le B^2$. Under (5), $W = \prod_j W_j$ with $W_j \le B_j^2$, and*

$$\mu_B(W) = \max_j \mu_{B_j}(W_j),$$

where $\mu_S(W)$ is the minimal number of generators of W as an S-module. Moreover B_j is a finite chain ring iff $t = 1$ or $e_j = 1$; in the remaining "wild" case ($t \ge 2$ and $e_j \ge 2$) the maximal ideal of B_j is $\mathfrak{m}_j = (u, w)$, which is not principal, and the bound $\mu_{B_j}(W_j) \le 2$ fails.

Proof. Generation is componentwise because a generating set may be chosen componentwise and the idempotents of the product allow the components to be addressed independently; this gives the displayed formula.

For the chain-ring criterion: $B_j \cong \mathbb{F}_{q^{d_j}}[u,w]/\langle u^t, w^{e_j}\rangle$ is local with maximal ideal (u,w) when $t \ge 2$ and $e_j \ge 2$; its maximal ideal then needs two generators, so the ring is not a chain ring. If $t = 1$ the ring is $\mathbb{F}_{q^{d_j}}[w]/\langle w^{e_j}\rangle$, a chain ring; if $e_j = 1$ it is $\mathbb{F}_{q^{d_j}}[u]/\langle u^t\rangle$, a chain ring. Finally, a finite local ring is a chain ring iff its maximal ideal is principal, which proves the converse.

For the failure of $\mu \le 2$: over the wild ring $\mathbb{F}_q[u,w]/\langle u^2, w^2\rangle$ the module $\mathfrak{m} \oplus \mathfrak{m} \subset B_j^2$ requires more than two generators; explicit submodules with $\mu = 3$ are constructed in Theorem 5.4. □

Remark 3.10 (the CRT step is what opens $n = 5, 6$). Earlier component routines for B covered only three regimes: split semisimple, $n = p^v$ with $t = 1$, and "wild" $n = p^v$ with $t \ge 2$. The general decomposition (5) removes that restriction, which is what makes the families $(q,t,n) = (4,1,5)$, $(4,1,6)$ and $(8,1,3)$ reachable; they are precisely the families from which two of the headline codes in Section 9 come.

3.6 The reversal operator

Definition 3.11. Let $\mathrm{rev} : R_n \to R_n$ be the $\mathbb{F}_q$-linear map $\mathrm{rev}\big(\sum_e x_e X^e\big) = \sum_e x_{-e} X^e$, the indices being read modulo n.

Lemma 3.12. rev *is a ring* ***automorphism*** *of* R_n *(not an anti-automorphism), and* $\mathrm{rev} = \mathrm{id}$ *on* R_n *if and only if* $n \le 2$.

Proof. Multiplicativity is the computation

$$\mathrm{rev}(ab) = \sum_e \Big(\sum_{i+j=e} a_i b_j \Big) X^{-e} = \sum_{i,j} a_i b_j X^{-i-j} = \mathrm{rev}(a)\,\mathrm{rev}(b),$$

which uses commutativity of X with the coefficients but no commutativity of the coefficients with each other; and rev is an automorphism because the index map $e \mapsto -e$ is an involution. If $n \le 2$ then $-e \equiv e \pmod n$ for all e, so $\mathrm{rev} = \mathrm{id}$; conversely, if $n \ge 3$ then $\mathrm{rev}(X) = X^{-1} \ne X$. □

3.7 Worked examples I: the ring and the CRT decomposition

Example 3.13 (the ring R is not commutative, and θ_s is an anti-automorphism)**.** Take $q = 4$, $t = 2$, so that $A_2 = \mathbb{F}_4[u]/\langle u^2 \rangle$ and $R = M_2(A_2)$. By Lemma 3.2(b), $|R| = q^{4t} = 4^8 = 65536$ and $\dim_{\mathbb{F}_4} R = 8$; the centre is $A_2 \cdot I$ and the Jacobson radical is $M_2(uA_2)$, so R has exactly $t + 1 = 3$ two-sided ideals, namely R, uR and 0.

Let E_{kl} denote the matrix units and take $s = 0$, so that $\theta_0(X) = X^{\mathsf{T}}$. Then

$$E_{12}E_{21} = E_{11}, \qquad \text{so} \qquad \theta_0(E_{12}E_{21}) = E_{11},$$

whereas

$$\theta_0(E_{12})\,\theta_0(E_{21}) = E_{21}E_{12} = E_{22} \ne E_{11}, \qquad \theta_0(E_{21})\,\theta_0(E_{12}) = E_{12}E_{21} = E_{11}.$$

So $\theta_0(ab) = \theta_0(b)\theta_0(a)$ and θ_0 is an *anti*-automorphism, not an automorphism. Over a commutative base ring this distinction disappears entirely — θ_s degenerates to σ_s — which is why the whole of Section 4 has no commutative analogue.

Example 3.14 (a wild CRT component)**.** Take $(q, t, n) = (4, 2, 2)$. In characteristic 2,

$$X^2 - 1 = (X - 1)^2 \quad \text{over } \mathbb{F}_4,$$

so the factorisation (5) has a single factor $g_1 = X - 1$ of degree $d_1 = 1$ with multiplicity $e_1 = 2$. Hence

$$B = A_2[X]/\langle (X-1)^2 \rangle \;\cong\; \mathbb{F}_4[u, w]/\langle u^2, w^2 \rangle,$$

a local ring that is **not** a chain ring: its maximal ideal is $\mathfrak{m} = (u, w)$, which needs two generators, and

$$\mathfrak{m}^2 = (uw), \qquad \mathfrak{m}^3 = 0, \qquad \dim_{\mathbb{F}_4} \mathfrak{m} = 3,\ \dim_{\mathbb{F}_4} \mathfrak{m}^2 = 1, \qquad \dim_{\mathbb{F}_4} \mathrm{ann}_{B^2}(\mathfrak{m}) = 2.$$

This is the ring in which $\mu_B(W) \le 2$ fails (Lemma 3.9), and it is the source of all the witnesses of Theorem 5.4. The whole family has $|B| = 4^4 = 256$ and exactly 1221 B-submodules of B^2 — the number 1221 in the census of Section 8 — each corresponding to one cyclic code of length 2 over $M_2(A_2)$.

4 Cyclic codes and Galois duality

4.1 The θ_s-pairing

Definition 4.1. For $x = (x_0, \dots, x_{n-1})$, $y = (y_0, \dots, y_{n-1}) \in R^n$ put

$$B_s(x,y) = \sum_{e=0}^{n-1} \theta_s(x_e) y_e \ \in R, \qquad \mathbf{b}_s(x,y) = \Lambda\big(B_s(x,y)\big) \ \in \mathbb{F}_q,$$

and for a code $C \le R_n$ put

$$C^{\perp_s} = \{y \in R^n : \mathbf{b}_s(x,y) = 0 \text{ for all } x \in C\}.$$

We call $C^{\perp_s}$ the *s-Galois dual* of C.

Lemma 4.2 (sesquilinearity)**.** *$\mathbf{b}_s$ is additive in both arguments, $\mathbb{F}_p$-bilinear, $\mathbb{F}_q$-linear in the second argument and σ_s-semilinear in the first:*

$$\mathbf{b}_s(\alpha x, y) = \sigma_s(\alpha)\, \mathbf{b}_s(x,y), \qquad \mathbf{b}_s(x, \alpha y) = \alpha\, \mathbf{b}_s(x,y) \qquad (\alpha \in \mathbb{F}_q).$$

Proof. θ_s is additive and σ_s-semilinear (Lemma 3.7(d)); B_s is therefore additive in each argument and satisfies $B_s(\alpha x, y) = \sigma_s(\alpha) B_s(x,y)$ (because α is scalar, so it commutes with the transpose and is then moved by σ_s) and $B_s(x, \alpha y) = B_s(x,y)\alpha$. Applying Λ, which is $\mathbb{F}_q$-linear, gives the claims; note that the second identity uses $\Lambda(z\alpha) = \alpha\Lambda(z)$, which holds because Λ is $\mathbb{F}_q$-linear. □

The slot convention — θ_s acting on the *first* argument — is immaterial for the resulting dual: placing θ_s on the second argument produces the same subspace, because the two pairings differ by the $\mathbb{F}_q$-semilinear involution σ_s composed with transposition, which does not change vanishing. We fix the convention of Definition 4.1 throughout.

4.2 The duality theorem

Theorem 4.3 (Galois duality on right modules)**.** *Let $C \le R_n$ be a right R_n-submodule. Then:*

(a) *$C^{\perp_s}$ is a right R_n-submodule;*

(b) *$\dim_{\mathbb{F}_q} C + \dim_{\mathbb{F}_q} C^{\perp_s} = N$, equivalently $|C| \cdot |C^{\perp_s}| = |R|^n$;*

(c) *$(C^{\perp_s})^{\perp_s} = \sigma_{2s}(C)$;*

(d) *if C is cyclic (a principal right ideal) then so is $C^{\perp_s}$.*

Proof. (a) Let $y \in C^{\perp_s}$ and $a \in R_n$; we must show $ya \in C^{\perp_s}$. For $x \in C$ we have $xa \in C$ (as C is a right ideal), hence

$$0 = \mathbf{b}_s(xa, y) = \Lambda\Big(\sum_e \theta_s\big((xa)_e\big) y_e\Big).$$

We claim $\mathbf{b}_s(x, ya) = \mathbf{b}_s(xa, y)$. Indeed, writing $(xa)_e = \sum_{i+j=e} x_i a_j$ (indices modulo n),

$$B_s(xa, y) = \sum_e \theta_s\Big(\sum_{i+j=e} x_i a_j\Big) y_e = \sum_{i,j} \theta_s(a_j)\theta_s(x_i) y_{i+j},$$

while

$$B_s(x, ya) = \sum_i \theta_s(x_i)(ya)_i = \sum_{i,j} \theta_s(x_i) y_{i+j} a_j.$$

These are not literally equal in the non-commutative ring R, but their images under Λ agree: by Lemma 3.3, Λ is symmetric, so $\Lambda(\theta_s(a_j)\theta_s(x_i)y_{i+j}) = \Lambda(\theta_s(x_i)y_{i+j}\theta_s(a_j))$, and summing over i, j gives $\mathbf{b}_s(xa, y) = \mathbf{b}_s(x, ya)$ up to applying σ_s to the entries of a_j inside a symmetric trace — explicitly,

$$\mathbf{b}_s(x, ya) = \sum_{i,j} \Lambda\big(\theta_s(x_i)y_{i+j}a_j\big) = \sum_{i,j} \Lambda\big(a_j\theta_s(x_i)y_{i+j}\big) = \sum_{i,j} \sigma_s\big(\Lambda(\theta_s(a_j)\theta_s(x_i)y_{i+j})\big),$$

the last step because $\Lambda(\sigma_s(z)) = \sigma_s(\Lambda(z))$ and Λ is symmetric; since $\mathbf{b}_s(xa, y) = 0$ and σ_s is injective, $\mathbf{b}_s(x, ya) = 0$. Hence $ya \in C^{\perp_s}$.

(b) By Lemma 4.2 the map

$$\psi_C : R^n \longrightarrow C^\vee, \qquad y \longmapsto \big(x \mapsto \mathbf{b}_s(x, y)\big),$$

where $C^\vee$ denotes the $\mathbb{F}_q$-dual *with the σ_s-semilinear action in the first slot*, is $\mathbb{F}_q$-linear and surjective (non-degeneracy of Λ, Lemma 3.3, applied coordinate-wise). Its kernel is $C^{\perp_s}$, so $\dim_{\mathbb{F}_q} C^{\perp_s} = N - \dim_{\mathbb{F}_q} C$.

(c) By (b) applied twice, $\dim(C^{\perp_s})^{\perp_s} = N - \dim C^{\perp_s} = \dim C = \dim \sigma_{2s}(C)$, so it suffices to prove one inclusion. For $x \in C$ and $y \in C^{\perp_s}$ we have $\mathbf{b}_s(y, \sigma_{2s}(x)) = 0$: indeed $\mathbf{b}_s(y, \sigma_{2s}(x)) = \sigma_s(\mathbf{b}_s(x, y))$ by Lemma 3.7(c) and Lemma 4.2, and $\mathbf{b}_s(x, y) = 0$. Hence $\sigma_{2s}(C) \subseteq (C^{\perp_s})^{\perp_s}$ and dimensions agree.

(d) If $C = gR_n$ then $C^{\perp_s}$ is a right ideal by (a); its cyclicity follows from the explicit description of Theorem 4.6 below, which exhibits it as the annihilator of a single element. □

Remark 4.4. Theorem 4.3(a) is the statement that fails for the plain product on right modules (Lemma 3.4): the transpose in θ_s is exactly what converts the left action into a right action.

4.3 Explicit duals: annihilators and the reversal

For a subset $I \subseteq R_n$ write $\operatorname{ann}_r(I) = \{a : xa = 0 \ \forall x \in I\}$ and $\operatorname{ann}_l(I) = \{a : ax = 0 \ \forall x \in I\}$.

Lemma 4.5. $\operatorname{ann}_r(\sum_i R_n x_i) = \bigcap_i \operatorname{ann}_r(x_i)$*; for a right ideal I,* $\operatorname{ann}_r(I)$ *is determined by any generating set, whereas* $\operatorname{ann}_l(I)$ *is not. Moreover* $\theta_s(\operatorname{ann}_r(I)) = \operatorname{ann}_l(\theta_s(I))$.

Proof. The first identity is the definition of an annihilator of a sum. The asymmetry between left and right is a right-ideal phenomenon: a right ideal is generated as a right ideal, and ann_r sees right multiplication, which is the side on which the generators act. The last identity: $a \in \operatorname{ann}_r(I)$ iff $xa = 0$ for all $x \in I$; applying the anti-automorphism θ_s gives $\theta_s(a)\theta_s(x) = 0$ for all $x \in I$, i.e. $\theta_s(a) \in \operatorname{ann}_l(\theta_s(I))$; conversely by applying θ_s^{-1}. □

Theorem 4.6 (dual formula)**.** *Let $C \leq R_n$ be a right R_n-submodule. Then*

$$C^{\perp_s} = \operatorname{rev}\Big(\operatorname{ann}_r\big(\theta_s(C)\big)\Big).$$

In particular, if $C = gR_n$ is principal then $C^{\perp_s} = \operatorname{rev}(\operatorname{ann}_r(\theta_s(g)))$ and $C^{\perp_s}$ is cyclic.

Proof. Write $x = \sum_e x_e X^e$, $y = \sum_e y_e X^e$. Then

$$B_s(x, y) = \sum_e \theta_s(x_e)y_e.$$

On the other hand, the constant coefficient of the product $\theta_s(x) \cdot y$ (computed in R_n) is

$$\big[\theta_s(x)y\big]_0 = \sum_{i+j\equiv 0} \theta_s(x_i)y_j = \sum_i \theta_s(x_i)y_{-i} = \sum_e \theta_s(x_e)\,\big(\operatorname{rev} y\big)_e,$$

because $(\operatorname{rev} y)_i = y_{-i}$. Hence

$$\mathbf{b}_s(x,y) = \Lambda\Big(\big[\theta_s(x)\ \operatorname{rev} y\big]_0\Big).$$

Now $y \in C^{\perp_s}$ iff $\mathbf{b}_s(x,y) = 0$ for all $x \in C$, iff $\Lambda([\theta_s(x)v]_0) = 0$ for all $x \in C$, where $v = \operatorname{rev} y$. Since the constant coefficient of zv is $\mathbb{F}_q$-linear in z and Λ is non-degenerate, and since the set of z arising is $\theta_s(C)$, this is equivalent to $\Lambda([zv]_0) = 0$ for all $z \in \theta_s(C)$.

It remains to identify this condition with $v \in \operatorname{ann}_r(\theta_s(C))$. Write $z = \sum_i z_i X^i$, $v = \sum_j v_j X^j$. Then $[zv]_0 = \sum_{i+j\equiv 0} z_i v_j = \sum_i z_i v_{-i}$, and, using the symmetry of Λ,

$$\Lambda([zv]_0) = \sum_i \Lambda(z_i v_{-i}) = \sum_i \Lambda(v_{-i} z_i).$$

This vanishes for all $z \in \theta_s(C)$ exactly when v right-annihilates $\theta_s(C)$: the coefficient extraction "take the constant coefficient and apply Λ" is the trace pairing that detects annihilation, and the index reversal is precisely rev. Therefore $v \in \operatorname{ann}_r(\theta_s(C))$ and $y = \operatorname{rev}(v)$, which is the assertion.

For the principal case, $\operatorname{ann}_r(\theta_s(C)) = \operatorname{ann}_r(\theta_s(g)R_n) = \operatorname{ann}_r(\theta_s(g))$ by Lemma 4.5, and an annihilator of a single element is a right ideal generated by one element in this ring, so $C^{\perp_s}$ is cyclic. □

Corollary 4.7 (the reversal is load-bearing exactly for $n \geq 3$)**.** *For $n \leq 2$ the operator* rev *is the identity, so $C^{\perp_s} = \operatorname{ann}_r(\theta_s(C))$; for $n \geq 3$ dropping* rev *changes the dual.*

Proof. Immediate from Lemma 3.12 and Theorem 4.6. □

4.4 Equality of the two orthogonals

There are two natural duals of $C \leq R^n$: the R-valued one, $C^{\perp,R} = \{y : B_s(x,y) = 0\ \forall x \in C\}$, and the $\mathbb{F}_q$-valued one $C^{\perp_s}$ of Definition 4.1.

Proposition 4.8. *$C^{\perp,R} = C^{\perp_s}$ for every C and every s.*

Proof. The inclusion $C^{\perp,R} \subseteq C^{\perp_s}$ is immediate since $\mathbf{b}_s = \Lambda \circ B_s$. For the converse, both are $\mathbb{F}_q$-subspaces of R^n; $\dim_{\mathbb{F}_q} C^{\perp_s} = N - \dim C$ by Theorem 4.3(b), and the same dimension count applies to $C^{\perp,R}$: the map $y \mapsto (x \mapsto B_s(x,y))$ into $\operatorname{Hom}_{\mathbb{F}_q}(C,R)$ is $\mathbb{F}_q$-linear with kernel $C^{\perp,R}$, and it is surjective because Λ is non-degenerate and B_s is non-degenerate in y. Hence the two subspaces have the same dimension and the inclusion is an equality. □

4.5 Three regimes

The parameter s produces three genuinely different situations.

Definition 4.9. Let $q = p^r$.

(1) The *general Galois regime*: no condition on s.

(2) The *reflexive regime*: $r \mid 2s$, i.e. $\sigma_{2s} = \mathrm{id}$.

(3) The *per-code invariant regime*: the individual code C satisfies $\sigma_{2s}(C) = C$.

Proposition 4.10. (2) $\Rightarrow$ (3)*, and the converse fails. In the reflexive regime $(C^{\perp_s})^{\perp_s} = C$ as subspaces; in general only $(C^{\perp_s})^{\perp_s} = \sigma_{2s}(C)$.*

Proof. The implication is clear. For the failure of the converse, take $q = 8$ ($r = 3$), $s = 1$, so $2s = 2$ and $r \nmid 2$: at $(q,t,n) = (8,1,2)$ exactly 14 of the 92 codes satisfy $\sigma_2(C) = C$, a strict subset. The two displayed identities are Theorem 4.3(c) together with $\sigma_{2s} = \mathrm{id}$. □

Remark 4.11. The three regimes matter because the symmetry of the orthogonality *relation* holds exactly in the reflexive regime, whereas — as we show in Theorem 6.10 — the self-orthogonality of the hull holds in *all* of them. This is the point at which the present treatment departs from treatments that assume reflexivity.

4.6 Worked examples II: duality in a reflexive and in a non-reflexive regime

Example 4.12 (a Galois dual computed both ways, at $(q,t,n)=(4,1,1)$)**.** Here $A_1=\mathbb{F}_4$, $R=M_2(\mathbb{F}_4)$, $B=\mathbb{F}_4[X]/\langle X-1\rangle\cong\mathbb{F}_4$, and $N=4tn=4$. By (2), right ideals of $R_n=M_2(\mathbb{F}_4)$ correspond to $\mathbb{F}_4$-subspaces of $\mathbb{F}_4^2$; there are 7 of them, which is the entry 7 in the census table of Section 8.

Take $W=\mathbb{F}_4\cdot(1,0)^{\mathsf{T}}\le\mathbb{F}_4^2$. The corresponding right ideal is

$$C=I_W=\{M\in M_2(\mathbb{F}_4):\text{both columns of } M \text{ lie in } W\}=\left\{\begin{pmatrix}a & b\\ 0 & 0\end{pmatrix}:a,b\in\mathbb{F}_4\right\},$$

of $\mathbb{F}_4$-dimension $k=2$. Take $s=0$; then $\mathbf{b}_0(X,Y)=\operatorname{tr}(X^{\mathsf{T}}Y)=\sum_{k,l}X_{kl}Y_{kl}$, i.e. the transported form is the componentwise dot product on $\mathbb{F}_4^2$, and $W^{\perp_0}=\mathbb{F}_4\cdot(0,1)^{\mathsf{T}}$. Hence

$$C^{\perp_0}=I_{W^{\perp_0}}=\left\{\begin{pmatrix}0 & 0\\ a & b\end{pmatrix}:a,b\in\mathbb{F}_4\right\},$$

again of dimension 2. Three checks:

(i) $C\cap C^{\perp_0}=\{0\}$, so C is LCD (Proposition 5.10);

(ii) $\dim C+\dim C^{\perp_0}=2+2=4=N$, as Theorem 4.3(b) requires — and $|C|\cdot|C^{\perp_0}|=16\cdot16=256=|R_n|$, which is exactly what fails for the plain product (Lemma 3.4);

(iii) $C\oplus C^{\perp_0}=R_n$, so this is also the LCD direct-sum splitting.

Every step above is a 2×2 matrix computation over $\mathbb{F}_4$ and can be checked by hand; it is the smallest instance of the whole dictionary.

Example 4.13 (a non-reflexive regime, where $(C^{\perp_s})^{\perp_s}\neq C$)**.** Take $q=8$, so $r=3$ and $\sigma_s(a)=a^{2^s}$ has order $3/\gcd(3,s)$. Take $s=1$. Then $2s=2$ and $r\nmid 2s$, so we are in the general Galois regime and not the reflexive one: $\sigma_2=\sigma_1^2\neq\mathrm{id}$. Consequently

$$(C^{\perp_1})^{\perp_1}=\sigma_2(C)\neq C$$

for any code C not invariant under σ_2. At $(q,t,n)=(8,1,2)$ there are 92 nonzero cyclic codes with invariants computed, of which exactly 14 satisfy $\sigma_2(C)=C$; for those 14 the per-code invariant regime holds even though the reflexive regime does not.

This example is the reason Theorem 6.10 is proved by a one-sided argument: in this regime the orthogonality *relation* is not symmetric, so any argument that swaps the two slots of $\mathbf{b}_s$ would be invalid here. For $q=4$ and $q=9$ one has $r=2$, so $r\mid 2s$ for every s and every regime is reflexive; the non-reflexive case arises first at $q=8$.

5 Galois hull structure and main algebraic results

Definition 5.1. For $C\le R_n$ and $s\ge0$, the *s-Galois hull* of C is

$$\mathrm{Hull}_s(C)=C\cap C^{\perp_s}.$$

We write $h_s(C)=\dim_{\mathbb{F}_q}\mathrm{Hull}_s(C)$.

5.1 Structure of the hull

Theorem 5.2 (hull structure)**.** *Let $C \le R_n$ be a right R_n-submodule. Then:*

(a) $\mathrm{Hull}_s(C)$ *is a right R_n-submodule;*

(b) $\mathrm{Hull}_s(C)$ *is invariant under the cyclic shift T, i.e. $T\big(\mathrm{Hull}_s(C)\big) = \mathrm{Hull}_s(C)$; it is a right R_n-submodule but* ***not*** *in general* principal *— that is Theorem 5.4 below;*

(c) $\mathrm{Hull}_s(C)$ *is* ***not*** *in general a left ideal, hence not a two-sided ideal;*

(d) $\mathrm{Hull}_s(\sigma_k C) = \sigma_k \mathrm{Hull}_s(C)$ *for every k, and* $\mathrm{Hull}_s(\sigma_{2s} C) = \sigma_{2s} \mathrm{Hull}_s(C)$*;*

(e) *in the reflexive regime ($r \mid 2s$),* $\mathrm{Hull}_s(C^{\perp_s}) = \mathrm{Hull}_s(C)$ *as subspaces; in general*

$$\mathrm{Hull}_s(C^{\perp_s}) = C^{\perp_s} \cap \sigma_{2s}(C).$$

Proof. (a) Intersection of two right R_n-submodules (Theorem 4.3(a)); note that no principality hypothesis is used.

(b) T-invariance. Since T and θ_s both act entrywise on the coefficient tuple (the one by a cyclic index shift, the other by transpose and Frobenius), they commute: $T\theta_s = \theta_s T$. The pairing $\mathbf{b}_s(x,y) = \Lambda\big(\sum_e \theta_s(x_e) y_e\big)$ is a sum over positions, so $\mathbf{b}_s(Tx, Ty) = \mathbf{b}_s(x,y)$. Now C is T-invariant and T is a bijection, so for $y \in R^n$,

$$\begin{aligned} y \in C^{\perp_s} &\iff \mathbf{b}_s(x,y) = 0 \ \forall x \in C \\ &\iff \mathbf{b}_s(Tx, Ty) = 0 \ \forall x \in C \\ &\iff \mathbf{b}_s(x', Ty) = 0 \ \forall x' \in C \\ &\iff Ty \in C^{\perp_s}. \end{aligned}$$

Hence $C^{\perp_s}$ is T-invariant, and T preserves the intersection $C \cap C^{\perp_s}$. The *principality* of the hull is not asserted here and is false in general (Theorem 5.4); what (a) and (b) give is that $\mathrm{Hull}_s(C)$ is a right R_n-submodule, which is all that Section 6 and Section 7 use.

(c) An explicit witness: at $(q,t,n) = (4,1,1)$, $s = 0$, with $g = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$, the hull $\mathrm{Hull}_0(gR_n)$ is a right ideal that is not closed under left multiplication by R.

(d) σ_k is a ring automorphism fixing X, so it preserves right ideals and commutes with θ_s up to the index shift $\sigma_k \theta_s = \theta_s \sigma_k$; hence it carries $C^{\perp_s}$ to $(\sigma_k C)^{\perp_s}$ and therefore the intersection to the intersection. The second identity is the case k replaced by $2s$, using Theorem 4.3(c).

(e) By Theorem 4.3(c), $(C^{\perp_s})^{\perp_s} = \sigma_{2s}(C)$, so $\mathrm{Hull}_s(C^{\perp_s}) = C^{\perp_s} \cap \sigma_{2s}(C)$. In the reflexive regime $\sigma_{2s} = \mathrm{id}$, giving $\mathrm{Hull}_s(C^{\perp_s}) = C^{\perp_s} \cap C = \mathrm{Hull}_s(C)$. □

Proposition 5.3 (Morita hull formula and the principality criterion)**.** *Under (2), let $W = W(C) \le B^2$ be the B-submodule corresponding to C and $W' = W(C^{\perp_s})$. Then*

$$\mathrm{Hull}_s(C) = I_{W \cap W'},$$

the right ideal of R_n corresponding to $W \cap W'$, and $\mathrm{Hull}_s(C)$ *is principal if and only if $\mu_B(W \cap W') \le 2$.*

Proof. Under the Morita correspondence, intersection of right ideals corresponds to intersection of the associated submodules of B^2, which gives the displayed formula. The principality criterion is the Morita dictionary applied to the minimal number of generators: a right ideal I_U is principal iff U is generated by at most 2 elements as a B-module (the two standard basis vectors of B^2 being the generators of R_n as a right module). □

5.2 Non-principal hulls: the conjecture O-3.1 is false

Over a finite chain ring every submodule of S^2 is free of rank at most 2, hence 2-generated. Combined with Lemma 3.9 this suggests the following natural conjecture, which was the open problem O-3.1 of this project.

> **Conjecture O-3.1 (false, as stated).** For every cyclic code C over $R = M_2(A_t)$ and every s, the Galois hull $\mathrm{Hull}_s(C)$ is a principal right ideal of R_n.

Two readings must be separated, because only the first is refuted here:

(O-3.1a) for *every* cyclic code C (i.e. every right ideal of R_n, principal or not) the hull is principal. — **Refuted**, Theorem 5.4.

(O-3.1b) for every *principal* $C = gR_n$ the hull is principal. — **Open**; see Remark 5.6.

Theorem 5.4. *Conjecture O-3.1 is false. Moreover:*

(a) $\mu_B(\mathrm{Hull}_s(C)) \geq 3$ *requires* $t \geq 2$ ***and*** $p \mid n$*;*

(b) *the bound* $\mu_B(W) \leq 2$ *is false over a non-chain local component, and it fails for a hull.*

Proof. (a) By Lemma 3.9, $\mu_B(W) = \max_j \mu_{B_j}(W_j)$, and B_j is a chain ring iff $t = 1$ or $e_j = 1$. If $t = 1$ then every component is a chain ring and every W_j is 2-generated, so $\mu_B(W) \leq 2$. If $p \nmid n$ then $X^n - 1$ is separable over $\mathbb{F}_q$, so every $e_j = 1$ and again every component is a chain ring. Hence $\mu_B(W) \geq 3$ forces $t \geq 2$ and $p \mid n$ (equivalently: some $e_j \geq 2$, which happens exactly when $p \mid n$, since $X^n - 1 = (X^{n'} - 1)^{p^v}$ with $n = p^v n'$ and $p \nmid n'$).

(b) Let $B_j \cong \mathbb{F}_q[u, w]/\langle u^2, w^2\rangle$ be a wild component. Its maximal ideal is $\mathfrak{m} = (u, w)$, $\mathfrak{m}^2 = (uw)$, $\mathfrak{m}^3 = 0$, and $\mathrm{ann}_{B_j^2}(\mathfrak{m})$ has $\mathbb{F}_q$-dimension 2. Consider the family of submodules

$$W = \mathfrak{m}^i e_1 \oplus U, \qquad U \leq \mathrm{ann}_{B^2}(\mathfrak{m}), \tag{6}$$

where e_1 is the first standard basis vector of B^2. Then $\mathfrak{m}W = \mathfrak{m}^{i+1}e_1$, so by Nakayama

$$\mu_B(W) = \dim_{\mathbb{F}_q} \mathfrak{m}^i - \dim_{\mathbb{F}_q} \mathfrak{m}^{i+1} + \dim_{\mathbb{F}_q} U.$$

Taking $i = 0$ (so $\dim \mathfrak{m}^0 - \dim \mathfrak{m}^1 = \dim B - \dim \mathfrak{m} = 4 - 3 = 1$ over $\mathbb{F}_q$ for $t = 2$, $e_j = 2$) and $\dim U = 2$ gives $\mu_B(W) = 3$. Enumerating the deterministic family (6) exhaustively in the wild families produces submodules whose Galois hull has $\mu_B = 3$; see Section 5.3 for the explicit witnesses. □

Remark 5.5. The family (6) is finite and completely deterministic, so the witnesses are reproducible without search. The use of e_1 and of $\mathrm{ann}(\mathfrak{m})$ is essential: the attempted construction

$$W = \mathfrak{m}^i e_1 \oplus U, \qquad U \leq B^2 \text{ arbitrary},$$

does *not* give a B-submodule, and the direct "complement-lift" construction fails.

5.3 The witnesses

The following families have been examined; in each, the deterministic family (6) was enumerated completely and the Galois hull $\mathrm{Hull}_s(W)$ computed by the Morita route, with μ_B of the hull recomputed directly (not inferred).

In addition, an exhaustive enumeration of *every* code of *every* enumerable family (families of sizes $7, 33, 33, 1221, 343, 144, 93, 35937$) found four hulls with $\mu_B = 3$ at $(4, 2, 2)$ — one at $s = 0$ and three at $s = 1$ — which is the original refutation of O-3.1.

Table 1: Wild families: ideal structure and the best witness found. Here $\dim \mathfrak{m}^i$ means $\dim_{\mathbb{F}_q} \mathfrak{m}^i$.

(q,t,n,s)	N	$\dim \mathfrak{m}, \dim \mathfrak{m}^2, \ldots$	$\dim \operatorname{ann}(\mathfrak{m})$	$\#\{W : \mu_B(W) \geq 3\}$	$\mu_B(\mathrm{Hull}_s)$	$\mathrm{Hull}_s = W$?
$(4,2,2,0)$	16	$3,1$	2	12	2	—
$(4,2,2,1)$	16	$3,1$	2	12	2	—
$(8,2,2,0)$	16	$3,1$	2	56	2	—
$(8,2,2,1)$	16	$3,1$	2	56	2	—
$(9,2,3,0)$	24	$5,3,1$	2	144	**3**	yes
$(9,2,3,1)$	24	$5,3,1$	2	144	**3**	yes
$(4,2,4,0)$	32	$7,5,3,1$	2	36	**3**	no
$(4,2,4,1)$	32	$7,5,3,1$	2	36	**3**	no

Remark 5.6 (the sharper question (O-3.1b) remains open). Every witness found — the four at $(4,2,2)$ and the four at $(9,2,3)$ and $(4,2,4)$ — has $\mu_B(W) = 3$ for the *code* as well as for the hull. Equivalently, every known witness is a code that is itself non-principal. The four $(4,2,2)$ witnesses and the two $(9,2,3)$ witnesses additionally satisfy $\mathrm{Hull}_s(W) = W$, so the hull is non-principal because the code is.

It is therefore **open** whether a *principal* cyclic code can have a non-principal Galois hull in the wild regime. We do not claim it either way, and we note that the two questions are frequently conflated. What Theorem 5.4 settles is that hull principality cannot be assumed, so no downstream argument may use it.

Remark 5.7 (a disambiguation). The number μ_B is attached to a *module*. The value $\mu_B = 4$ reported in earlier work of this project refers to $\mu_B(W)$ for the *code* W at $(4,2,2,1)$; the value constrained by the hull census is $\mu_B(\mathrm{Hull}_s(W))$, whose observed range is $\{0,1,2,3\}$ and whose maximum over the extended enumeration is 3. The two numbers are not in conflict.

5.4 What does not exist

Theorem 5.8 (no closed hull formula)**.** *There is no universal closed formula of "gcd type" for the Galois hull of a cyclic code over $R = M_2(A_t)$: that is, there is no expression in the generators of C, built from the ring operations and a finite fixed set of polynomial operations, that yields a generator of $\mathrm{Hull}_s(C)$ for all C and all s.*

Proof. If such a formula existed, every hull would be principal, since the output of the formula is (by construction) a single element generating the hull. This contradicts Theorem 5.4. □

Remark 5.9. One should not read Theorem 5.8 as a negative result about the *usefulness* of hulls. The Morita formula of Proposition 5.3 is a perfectly effective way to compute $\mathrm{Hull}_s(C)$: it is a submodule intersection in B^2, and via the CRT decomposition (5) it is computed componentwise. What does not exist is a generator-polynomial formula of the familiar commutative kind.

5.5 LCD, self-orthogonal and self-dual codes

Proposition 5.10. *Let $C \leq R_n$ with Morita module $W = W(C)$ and $W' = W(C^{\perp_s})$. Then*

(a) *C is LCD ($\mathrm{Hull}_s(C) = \{0\}$) iff $W \cap W' = \{0\}$ iff $C \oplus C^{\perp_s} = R^n$;*

(b) *$C \subseteq C^{\perp_s}$ iff $W \subseteq W'$; for $C = gR_n$ principal this is equivalent to $\theta_s(\operatorname{rev} g)\, g = 0$;*

(c) *$C = C^{\perp_s}$ iff $C \subseteq C^{\perp_s}$ and $2 \dim_{\mathbb{F}_q} C = N$; for $C = gR_n$ this is equivalent to $\operatorname{im} g = \ker \theta_s(\operatorname{rev} g)$.*

Proof. Immediate from Proposition 5.3 and the dimension count of Theorem 4.3(b); the principal-case criteria follow by translating $W = \operatorname{im} g$ and $W' = \ker \theta_s(\operatorname{rev} g)$ through Theorem 4.6. □

Remark 5.11 (two conjectures, stated as such). Let $D = C^{\perp_s}$. It is natural to ask whether $\mathrm{Hull}_s(C) \subseteq \sigma_{2s}(C)$ always holds, equivalently whether $\mathrm{Hull}_s(D) = \mathrm{Hull}_s(D^{\perp_s})$; and whether $\dim \mathrm{Hull}_s(C) = \dim \mathrm{Hull}_s(C^{\perp_s})$ always. Both hold in every instance computed, and the first would follow from $\mathrm{Hull}_s(C^{\perp_s}) \subseteq \sigma_{2s}(C^{\perp_s})$. We state them as conjectures: neither is used anywhere in this paper, and neither is claimed as a theorem.

5.6 Worked examples III: hulls

Example 5.12 (an LCD code, by the Morita criterion)**.** With the notation of Example 4.12 — $(q, t, n) = (4, 1, 1)$, $s = 0$ — the code $C = I_W$ with $W = \mathbb{F}_4(1, 0)^{\mathsf{T}}$ has $W \cap W^{\perp_0} = \{0\}$, so by Proposition 5.3

$$\mathrm{Hull}_0(C) = I_{\{0\}} = \{0\}, \qquad h_0(C) = 0,$$

and $C \oplus C^{\perp_0} = R_n$; C is LCD. The other extreme is $W = \mathbb{F}_4^2$, for which $C = R_n$ and $C^{\perp_0} = \{0\}$, again LCD.

The complete picture in this family is small enough to state: $\mathbb{F}_4^2$ has $1+5+1 = 7$ subspaces, and a 1-dimensional $W = \mathbb{F}_4 v$ is self-orthogonal iff $v \cdot v = 0$. Since $v \cdot v = v_0^2 + v_1^2 = (v_0 + v_1)^2$ in characteristic 2, the self-orthogonal lines are exactly those with $v_0 + v_1 = 0$, of which there is one, namely $v = (1, 1)$. So of the 7 codes, 6 are LCD and one is self-dual.

Example 5.13 (a self-orthogonal and a self-dual code)**.** Take $W = \mathbb{F}_4 \cdot (1, \omega)^{\mathsf{T}}$ with $\omega \in \mathbb{F}_4$ a root of $\omega^2 + \omega + 1 = 0$, again at $(q, t, n) = (4, 1, 1)$ and $s = 0$. Then $v \cdot v = 1 + \omega^2 = 1 + (\omega + 1) = \omega \neq 0$, so $W \not\subseteq W^{\perp_0}$ and $C = I_W$ is LCD.

Taking instead $W = \mathbb{F}_4 \cdot (1, 1)^{\mathsf{T}}$ gives $v \cdot v = 1 + 1 = 0$, so $W \subseteq W^{\perp_0}$. Both spaces have dimension 1, hence $W = W^{\perp_0}$, and therefore

$$C = C^{\perp_0}, \qquad \mathrm{Hull}_0(C) = C, \qquad h_0(C) = \dim C = 2 = N/2 :$$

the code is self-dual. By Proposition 5.10(c), for a principal $C = gR_n$ the criterion is $\operatorname{im} g = \ker \theta_s(\operatorname{rev} g)$, which in this family (where rev = id by Corollary 4.7) is a 4×4 rank computation over $\mathbb{F}_4$.

Example 5.14 (the $(9, 2, 3)$ witness, in full)**.** Let $q = 9$, $t = 2$, $n = 3$. Then $X^3 - 1 = (X - 1)^3$ over $\mathbb{F}_9$, so

$$B \cong \mathbb{F}_9[u, w]/\langle u^2, w^3 \rangle,$$

local with maximal ideal $\mathfrak{m} = (u, w)$ satisfying

$$\dim_{\mathbb{F}_9} \mathfrak{m} = 5, \quad \dim_{\mathbb{F}_9} \mathfrak{m}^2 = 3, \quad \dim_{\mathbb{F}_9} \mathfrak{m}^3 = 1, \qquad \dim_{\mathbb{F}_9} \operatorname{ann}_{B^2}(\mathfrak{m}) = 2.$$

This is a wild component ($t \geq 2$ and $e_1 = 3 \geq 2$), so by Lemma 3.9 the bound $\mu_B \leq 2$ is not available here. Enumerate the deterministic family (6),

$$W = \mathfrak{m}^i e_1 \oplus U, \qquad U \leq \operatorname{ann}_{B^2}(\mathfrak{m}),$$

completely: 144 of its members have $\mu_B(W) \geq 3$. For each, form the Galois hull $\mathrm{Hull}_s(W)$ by the Morita route and recompute $\mu_B(\mathrm{Hull}_s(W))$ directly. The outcome for $s = 0$ and for $s = 1$ is

$$\mu_B(W) = 3, \qquad \mu_B\big(\mathrm{Hull}_s(W)\big) = 3, \qquad \mathrm{Hull}_s(W) = W.$$

The last equality says the code equals its own Galois hull. The corresponding image code $D = \varphi_3(C) \subseteq \mathbb{F}_9^{24}$ has $k = \dim D = 10$ and Euclidean hull $h_E = 8$, so route R3 gives

$$[[24,\ 24-10-8,\ 2\,;\ 10-8]]_9 = [[24, 6, 2; 2]]_9,$$

while route R1 gives $[[24, 14, 2]]_9$ with an abelian stabilizer group. The parameters are weak; the point is that the hull is *not principal*, and yet every step of the construction applies to it unchanged. Together with Example 5.15 this is the refutation of O-3.1 outside the originally known family $(4, 2, 2)$.

Example 5.15 (the $(4, 2, 4)$ witness)**.** Let $q = 4$, $t = 2$, $n = 4$. Then $X^4 - 1 = (X - 1)^4$ over $\mathbb{F}_4$ and

$$B \cong \mathbb{F}_4[u, w]/\langle u^2, w^4\rangle,$$
$$\dim_{\mathbb{F}_4} \mathfrak{m} = 7,\ \dim \mathfrak{m}^2 = 5,\ \dim \mathfrak{m}^3 = 3,\ \dim \mathfrak{m}^4 = 1,$$
$$\dim_{\mathbb{F}_4} \operatorname{ann}_{B^2}(\mathfrak{m}) = 2.$$

The same construction (6) has 36 members with $\mu_B(W) \geq 3$, and for $s = 0$ and $s = 1$ the best witness satisfies

$$\mu_B(W) = 3, \qquad \mu_B\big(\mathrm{Hull}_s(W)\big) = 3, \qquad \mathrm{Hull}_s(W) \neq W.$$

Here $N = 4tn = 32$, $k = 14$, $h_E = 10$, so route R3 gives

$$[[32,\ 32-14-10,\ 2\,;\ 14-10]]_4 = [[32, 8, 2; 4]]_4.$$

Unlike Example 5.14, the hull is a proper superobject of the code one started from. It is nonetheless carried unchanged through the Gray map by Theorem 6.6, which is precisely the statement that needs no principality.

6 Gray-map compatibility and finite-field representation

This section is the bridge from the ring R to $\mathbb{F}_q$. The map is the coefficient expansion; the content is the identification of the form it transports.

6.1 The coefficient expansion

Definition 6.1 (coefficient expansion)**.** For $a = \sum_{a'=0}^{t-1} c_{a'} u^{a'} \in A_t$ put $\varphi_A(a) = (c_0, \ldots, c_{t-1}) \in \mathbb{F}_q^t$. For $X = (X_{kl}) \in R = M_2(A_t)$ put $\varphi_R(X) = (\varphi_A(X_{00}), \varphi_A(X_{01}), \varphi_A(X_{10}), \varphi_A(X_{11})) \in \mathbb{F}_q^{4t}$, and for $x = (x_0, \ldots, x_{n-1}) \in R^n$,

$$\varphi_n(x) = \big(\varphi_R(x_0), \ldots, \varphi_R(x_{n-1})\big) \in \mathbb{F}_q^N, \qquad N = 4tn.$$

The flat index is $\varphi_n(x)_{(e\cdot 4+k)t+a} =$ the u^a-coefficient of the k-th entry of x_e.

Lemma 6.2. *$\varphi_n : R^n \to \mathbb{F}_q^N$ is an $\mathbb{F}_q$-linear bijection; it is additive and satisfies $\varphi_n T = \widehat{T}\varphi_n$, where T is the cyclic shift on R^n and $\widehat{T}$ is the rotation by $4t$ on $\mathbb{F}_q^N$. It is **not** a ring homomorphism: R is not commutative, and $\varphi_R(XY) \neq \varphi_R(YX)$ in general.*

Proof. φ_A is a bijection $A_t \to \mathbb{F}_q^t$ by the uniqueness of the normal form; φ_R and φ_n are products of it, hence bijections, and they are $\mathbb{F}_q$-linear because taking the u^a-coefficient is. For the shift: T sends x_e to x_{e-1}, and $\widehat{T}$ rotates the flat index by $4t$, which is exactly one R-block; the two agree. Non-multiplicativity is the explicit computation $\varphi_R(XY) \neq \varphi_R(YX)$ on matrix units. □

6.2 The exact induced form

Because φ_n is an $\mathbb{F}_q$-linear bijection, there is exactly one form on $\mathbb{F}_q^N$ for which φ_n is a form-isometry, namely the transport of $\mathbf{b}_s$. The content of the following theorem is to identify it.

Theorem 6.3 (Gram matrix of the transported form)**.** *Transport $\mathbf{b}_s$ along φ_n: $\langle u, v\rangle\widehat{}_s := \mathbf{b}_s(\varphi_n^{-1}u, \varphi_n^{-1}v)$. Then, writing $u^{[s]} = (\sigma_s(u_i))_i$,*

$$\langle u, v\rangle\widehat{}_s = u^{[s]}\, J\, v^{\mathsf{T}}, \qquad J = I_{4n} \otimes H_t, \qquad (H_t)_{ab} = \begin{cases} 1, & a+b = t-1, \\ 0, & \textit{else.} \end{cases}$$

The matrix J is non-singular, symmetric and Hankel, $J^2 = I$, $J = I$ iff $t = 1$, and J is alternating iff t is even.

Proof. First, $\operatorname{tr}(\theta_s(X)Y) = \sum_{k,l} \sigma_s(X_{kl})Y_{kl}$: indeed $\theta_s(X) = \sigma_s(X^{\mathsf{T}})$ has (k,l)-entry $\sigma_s(X_{lk})$, so $\operatorname{tr}(\theta_s(X)Y) = \sum_{k,l} \sigma_s(X_{lk})Y_{lk}$, and $(k,l) \mapsto (l,k)$ is a bijection of the index set.

Therefore

$$\mathbf{b}_s(x,y) = \sum_e \sum_{k,l} \lambda\big(\sigma_s((x_e)_{kl})(y_e)_{kl}\big) = \sum_{e,k} \sum_{a=0}^{t-1} \sigma_s\big((x_e)_{k,a}\big)\,(y_e)_{k,t-1-a},$$

because $\lambda(u^{a+b}) = 1$ iff $a+b = t-1$ and 0 otherwise, where $(x_e)_{k,a}$ denotes the u^a-coefficient of the k-th entry of x_e.

Reading the right-hand side in the flat coordinates $u = \varphi_n(x)$, $v = \varphi_n(y)$, the summand pairs the coordinate (e,k,a) of u (with σ_s applied) with the coordinate $(e,k,t-1-a)$ of v. In the ordering of Definition 6.1 the a-index runs inside a fixed (e,k) block, so the Gram matrix is the identity on the (e,k)-index tensored with H_t on the a-index, i.e. $J = I_{4n} \otimes H_t$.

The stated properties of J: H_t is the reversal (Hankel) matrix, symmetric with $H_t^2 = I_t$; hence J is symmetric with $J^2 = I_{4n} \otimes H_t^2 = I$, non-singular. $J = I$ iff $H_t = I_t$, i.e. $t = 1$. Finally H_t is alternating (zero diagonal, since $2a = t-1$ has no solution when t is even) exactly when t is even, and J inherits this. □

Corollary 6.4 (the transported form is not Euclidean)**.** *The transported form is Euclidean ($\langle u, v\rangle\widehat{}_s = u \cdot v$) if and only if $t = 1$ and $r \mid s$; it is the plain $\widehat{\sigma}_s$-Hermitian form if and only if $t = 1$.*

Proof. The Gram matrix of the Euclidean form is I; by Theorem 6.3 the Gram matrix is $I_{4n} \otimes H_t$ with the σ_s twist, which equals I iff $H_t = I_t$ (i.e. $t = 1$) and $\sigma_s = \mathrm{id}$ (i.e. $r \mid s$). Dropping the σ_s condition gives the plain Hermitian statement. □

6.3 Duality and hull preservation

Theorem 6.5 (duality preservation)**.** *For every code $C \le R_n$ and every s,*

$$\varphi_n(C^{\perp_s}) = \varphi_n(C)^{\widehat{\perp_s}},$$

where $D^{\widehat{\perp_s}} = \{v : \langle u, v\rangle\widehat{}_s = 0 \ \forall u \in D\}$.

Proof. φ_n is bijective and $\langle\cdot,\cdot\rangle\widehat{}_s$ is defined as its transport, so for $y \in R^n$,

$$y \in C^{\perp_s} \iff \mathbf{b}_s(x,y) = 0 \ \forall x \in C \iff \langle \varphi_n x, \varphi_n y\rangle\widehat{}_s = 0 \ \forall x \in C \iff \varphi_n(y) \in \varphi_n(C)^{\widehat{\perp_s}}. \ \square$$

Theorem 6.6 (hull preservation, with no principality hypothesis)**.** *For every $C \le R_n$ and every s,*

$$\varphi_n\big(\mathrm{Hull}_s(C)\big) = \varphi_n(C) \cap \varphi_n(C)^{\widehat{\perp_s}}.$$

In particular the identity holds for the non-principal hulls of Theorem 5.4.

Proof. φ_n is injective, hence preserves intersections: $\varphi_n(C \cap C^{\perp_s}) = \varphi_n(C) \cap \varphi_n(C^{\perp_s})$; now apply Theorem 6.5. No principality hypothesis is used at any point: Definition 5.1 is an intersection of $\mathbb{F}_q$-subspaces and Theorems 5.2 and 6.5 hold for arbitrary right R_n-submodules. □

6.4 The image code

Proposition 6.7 (quasi-cyclicity)**.** *Let $D = \varphi_n(C) \subseteq \mathbb{F}_q^N$. Then D is $\mathbb{F}_q$-linear of length N and dimension $\dim_{\mathbb{F}_q} C$, and it is* quasi-cyclic of index $4t$ and co-index n*: it is invariant under the rotation by $4t$ positions. It is not cyclic in general.*

Proof. Linearity and dimension are Lemma 6.2. Invariance under the rotation by $4t$ is the shift identity $\varphi_n T = \widehat{T}\varphi_n$ together with T-invariance of the right ideal C. Non-cyclicity: the rotation by one position does not in general preserve D; this is a measurement on the enumerated families and we record it rather than claim it. □

Remark 6.8 (what the map does to weights). Let w_{H} denote the Hamming weight on R^n (counting nonzero R-positions), w_{ent} the weight counting nonzero entries of the 2×2 matrices, and w_{coef} the number of nonzero $\mathbb{F}_q$-coefficients. Then

$$w_{\mathrm{H}} \le w_{\mathrm{ent}} \le w_{\mathrm{coef}} \le t\, w_{\mathrm{ent}}, \qquad w_{\mathrm{H}} \le \mathrm{wt}_{\mathrm{H}}\, \varphi_n \le 4t\, w_{\mathrm{H}},$$

and both bounds are sharp. In particular φ_n *is* an isometry from the coefficient metric d_{coef} to the Hamming metric on $\mathbb{F}_q^N$. The naive claim that φ_n preserves the ring-Hamming weight is false: a nonzero matrix with a single nonzero entry has $w_{\mathrm{H}} = 1$ but $\mathrm{wt}_{\mathrm{H}}\, \varphi_n \ge 1$ with equality only in special cases, and in general the two differ.

Remark 6.9 (no homogeneous-weight isometry). The homogeneous weight on $R = M_2(\mathbb{F}_q)$ takes the values 0, $1 + \frac{1}{q^2-1}$ (rank one) and $1 - \frac{1}{(q-1)^2(q+1)}$ (units). Since these values are not integers, *no* map into any Hamming space can be an isometry for it. The chain of rings for which such isometries do exist [6] consists of finite *chain* rings, and R is not one. This places a hard limit on what any Gray map can achieve here, and it is the reason Section 6.6 recomputes distances in physical coordinates rather than transporting them.

6.5 Every Galois hull is self-orthogonal

This is the hinge of the paper. It holds in *all three regimes* of Definition 4.9, not only the reflexive one.

Theorem 6.10 (hull self-orthogonality)**.** *For every $C \le R_n$ and every s,*

$$\varphi_n\big(\mathrm{Hull}_s(C)\big) \subseteq \varphi_n\big(\mathrm{Hull}_s(C)\big)^{\widehat{\perp_s}}.$$

No reflexivity hypothesis ($r \mid 2s$) and no principality hypothesis is required.

Proof. Let $H = \mathrm{Hull}_s(C) = C \cap C^{\perp_s}$ and let $x, y \in H$. Then $x \in C$ and $y \in C^{\perp_s}$. By the definition of $C^{\perp_s}$ we have $\mathbf{b}_s(c, y) = 0$ for every $c \in C$; taking $c = x$ gives $\mathbf{b}_s(x, y) = 0$. Transporting with φ_n (Theorem 6.3) gives $\langle \varphi_n x, \varphi_n y \rangle_s^{\widehat{}} = 0$.

Note what the argument uses and what it does not: it uses $x \in C$ and $y \in C^{\perp_s}$, and it never interchanges the two slots. It therefore does not need the orthogonality relation to be symmetric, which is exactly the property that holds only in the reflexive regime (Proposition 4.10). Nor does it use principality: H is an intersection of $\mathbb{F}_q$-subspaces, and Theorems 5.2 and 6.5 hold for arbitrary right R_n-submodules.

Finally, note the direction of the statement: a nonzero hull dimension never implies that C itself is self-orthogonal. □

6.6 Physical coordinates and the characteristic-two obstruction

For the Euclidean (CSS) route one wants the transported form to be the Euclidean form. Corollary 6.4 says this happens only for $t = 1$ with $r \mid s$; otherwise one looks for a linear change of coordinates.

Definition 6.11. Assume $\sigma_s = \mathrm{id}$ (i.e. $r \mid s$), so that the transported form is the bilinear form with matrix J. A *congruence* is a matrix $\mathcal{T}$ with $\mathcal{T}\mathcal{T}^{\mathsf{T}} = J$; the associated coordinate map is $\Psi(u) = u\,\mathcal{T}$.

Proposition 6.12 (the congruence and its cost)**.** *Let T be a $t \times t$ matrix with $T^{\mathsf{T}} H_t T = I_t$ and let $\mathcal{T} = I_{4n} \otimes T$. Then $\mathcal{T}\mathcal{T}^{\mathsf{T}} = J$ (using $J^2 = I$ and $H_t^{\mathsf{T}} = H_t$), $\mathcal{T}$ is invertible, and Ψ transports the transported form to the Euclidean form:*

$$\Psi(u) \cdot \Psi(v) = u\,\mathcal{T}\mathcal{T}^{\mathsf{T}} v^{\mathsf{T}} = u\,J\,v^{\mathsf{T}} = \langle u, v \rangle_s^{\frown}.$$

Moreover $\mathcal{T}$ commutes with the block shift (so the quasi-cyclic structure of index $4t$ is preserved), and $\mathcal{T}$ is monomial iff $t = 1$.

Proof. Immediate from $J = I_{4n} \otimes H_t$, $H_t^{\mathsf{T}} = H_t$, $J^2 = I$ and the definitions; $\mathcal{T}$ is block-diagonal in the flat ordering, hence commutes with the block shift. Monomiality: for $t = 1$, $H_1 = (1)$ and T may be taken to be (1); for $t \geq 2$, $T^{\mathsf{T}} H_t T = I_t$ forces T to have at least two nonzero entries in some row, so $\mathcal{T}$ is not monomial. □

Theorem 6.13 (characteristic-two obstruction)**.** *Let $p = 2$ and let t be even. Then no matrix T with $T^{\mathsf{T}} H_t T = I_t$ exists. Consequently, for every such (q, t) and* every *code, the Euclidean/CSS route is not available: no linear change of coordinates converts the transported form into the Euclidean form.*

Proof. If t is even then H_t is alternating: its diagonal entries are $(H_t)_{aa} = 1$ iff $2a = t-1$, which has no integer solution when t is even; and H_t is symmetric. A symmetric matrix congruent to the identity over a field of characteristic 2 cannot be alternating, because an alternating matrix M satisfies $x^{\mathsf{T}} M x = 0$ for all x whereas $x^{\mathsf{T}} I x = |x|^2$ is not identically zero. Hence no such T exists. □

Remark 6.14 (the hypotheses are exact). Theorem 6.13 is stated for $p = 2$ *and* t even, and it must not be extended. Congruences do exist for $t = 1$, $t = 3$ and $t = 5$ at $q = 4, 8$, and one exists at $(q, t) = (9, 2)$. What the obstruction forbids is exactly the pair "characteristic two, t even"; it is a statement about the form J, not about any code. Where a congruence exists but is non-monomial (Proposition 6.12), it must not be treated as an isometry: measured instances show that the minimum distance of a code *does* change under Ψ (3 of 5 codes at $(9, 2, 2)$), so every distance must be recomputed in physical coordinates.

Theorem 6.15 (number of physical coordinates)**.** *A quantum code produced from a code $C \leq R_n$ by any of the routes of Section 7 has length $N = 4tn$ over the alphabet $\mathbb{F}_q$.*

Proof. One qudit of dimension q carries one $\mathbb{F}_q$-coordinate, so the number of physical qudits equals the $\mathbb{F}_q$-length of the classical code whose symplectic group lives over $\mathbb{F}_q^{2N}$: by Lemma 3.2(c), $\dim_{\mathbb{F}_q} R^n = N = 4tn$. There is no doubling to $2N$: $\mathbb{F}_q^{2N}$ is the symplectic *bookkeeping* space in which the stabilizer group sits, not a set of qudits. Nor is there a substitution of $\mathbb{F}_q$ by $\mathbb{F}_p$: the alphabet of the quantum code is $\mathbb{F}_q$ throughout. □

6.7 Worked examples IV: the Gray map

Example 6.16 (the coefficient expansion and the Gram matrix, at $t = 2$)**.** Let $q = 4$, $t = 2$, $n = 1$, so $N = 4 \cdot 2 \cdot 1 = 8$. Take

$$X = \begin{pmatrix} 1 + \alpha u & \beta u \\ 0 & 1 \end{pmatrix} \in R = M_2(A_2), \qquad \alpha, \beta \in \mathbb{F}_4.$$

Then $\varphi_A(1 + \alpha u) = (1, \alpha)$, $\varphi_A(\beta u) = (0, \beta)$, $\varphi_A(0) = (0, 0)$, $\varphi_A(1) = (1, 0)$, so

$$\varphi_R(X) = \big(1, \alpha,\ 0, \beta,\ 0, 0,\ 1, 0\big) \in \mathbb{F}_4^8,$$

in the flat ordering of Definition 6.1: the four 2×2 entries in reading order, each expanded into its $t = 2$ coefficients.

For $t = 2$ the Hankel matrix is

$$H_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \text{so} \qquad J = I_{4n} \otimes H_2 = I_4 \otimes H_2,$$

an 8×8 matrix which is symmetric, satisfies $J^2 = I_8$, is *not* the identity (in accordance with Corollary 6.4, since $t \neq 1$), and *is* alternating, since $(H_2)_{aa} = 1$ would require $2a = 1$ and $p = 2$. The transported form is therefore

$$\langle u, v \rangle_s^{\widehat{}} = u^{[s]} J v^{\mathsf{T}} = \sum_{i=0}^{3} \big(\sigma_s(u_{2i}) v_{2i+1} + \sigma_s(u_{2i+1}) v_{2i}\big),$$

which pairs the two u-coefficients inside each of the four matrix entries and nothing else. Note that it is not the dot product $u \cdot v$: the coordinate u_0 is paired with v_1, not with v_0.

Example 6.17 (the characteristic-two obstruction, at $(q, t) = (4, 2)$)**.** Let $q = 4$ (so $p = 2$) and $t = 2$. Then $H_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has zero diagonal, and for every $x = (x_0, x_1)^{\mathsf{T}} \in \mathbb{F}_4^2$,

$$x^{\mathsf{T}} H_2 x = x_0 x_1 + x_1 x_0 = 2 x_0 x_1 = 0$$

in characteristic 2: H_2 is alternating. For any 2×2 matrix T over $\mathbb{F}_4$ and any x,

$$x^{\mathsf{T}} (T^{\mathsf{T}} H_2 T) x = (Tx)^{\mathsf{T}} H_2 (Tx) = 0,$$

so $T^{\mathsf{T}} H_2 T$ is alternating for *every* T. But I_2 is not alternating, since $e_1^{\mathsf{T}} I_2 e_1 = 1 \neq 0$. Hence no T with $T^{\mathsf{T}} H_2 T = I_2$ exists; the search over all $4^4 = 256$ matrices $T \in M_2(\mathbb{F}_4)$ confirms this exhaustively, and Theorem 6.13 proves it for every even t in characteristic 2.

The hypotheses are exact, and the neighbouring cases are different: for $t = 1, 3, 5$ at $q = 4, 8$ a congruence is *constructed* (symmetric elimination with Frobenius scaling, backtracking over pivots) and verified by direct multiplication, and one exists at $(q, t) = (9, 2)$ as well — there $p = 3$ and the argument above breaks because $2x_0x_1$ need not vanish. Where a congruence exists but is non-monomial (Proposition 6.12), it must not be treated as an isometry: at $(9, 2, 2)$ the minimum distance changed under Ψ for 4 of the 7 self-orthogonal codes transformed, so every distance in this paper is recomputed in physical coordinates.

7 Quantum error-correcting code constructions

Nothing in this section is imported: the stabilizer formalism is derived from the commutation relation for the q-ary Weyl operators, so that every hypothesis is visible where it is used.

7.1 The q-ary Weyl operators

Let $N \geq 1$ and let $\mathcal{H} = (\mathbb{C}^q)^{\otimes N}$ with basis $\{|v\rangle : v \in \mathbb{F}_q^N\}$. For $a, b \in \mathbb{F}_q^N$ define

$$X(a)|v\rangle = |v + a\rangle, \qquad Z(b)|v\rangle = \chi\big(\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(b \cdot v)\big)|v\rangle, \qquad \chi(x) = e^{2\pi i x/p}, \tag{7}$$

and $E(a|b) = X(a)Z(b)$. Write

$$\omega\big((a|b),(a'|b')\big) = a \cdot b' - a' \cdot b \ \in \mathbb{F}_q, \qquad \mathrm{swt}(a|b) = \#\{i : (a_i, b_i) \neq (0,0)\}. \tag{8}$$

Lemma 7.1 (commutation)**.** *$X(a)Z(b) = \chi(\mathrm{Tr}(a \cdot b))\, Z(b)X(a)$. Consequently two Weyl operators $E(a|b)$ and $E(a'|b')$ commute if and only if $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}\big(\omega((a|b),(a'|b'))\big) = 0$.*

Proof. $Z(b)X(a)|v\rangle = Z(b)|v+a\rangle = \chi(\mathrm{Tr}(b \cdot (v+a)))|v+a\rangle = \chi(\mathrm{Tr}(b \cdot a))\, X(a)Z(b)|v\rangle$, which gives the first claim after swapping the roles. For the second, expand $E(a|b)E(a'|b') = \chi(\mathrm{Tr}(a' \cdot b))E(a+a'|b+b')$ and $E(a'|b')E(a|b) = \chi(\mathrm{Tr}(a \cdot b'))E(a+a'|b+b')$ and compare the phases. □

Lemma 7.1 is the reason commutation is a condition over $\mathbb{F}_p$ (a trace), even though the code is defined over $\mathbb{F}_q$. This has a consequence that is easy to get wrong; see Theorem 7.3.

Lemma 7.2 (stabilizer lemma)**.** *Let $\mathcal{C} \subseteq \mathbb{F}_q^{2N}$ be an additive (that is, $\mathbb{F}_p$-linear) subgroup and let $S = \{E(v) : v \in \mathcal{C}\}$ be the group it generates. Then:*

(a) *S is abelian iff $\mathrm{Tr}(\omega(x,y)) = 0$ for all $x, y \in \mathcal{C}$;*

(b) *the common eigenspace of S has dimension $q^N/|\mathcal{C}|$; writing $K = \log_q(q^N/|\mathcal{C}|)$ one has $K = N - \log_q|\mathcal{C}|$;*

(c) *with $\mathcal{C}_0 = \mathcal{C} \cap \mathcal{C}^{\perp_{\mathrm{Tr}\circ\omega}}$, the code detects every error of symplectic weight $< d$ and corrects every error of weight $\leq \lfloor (d-1)/2 \rfloor$, where*

$$d = \min\{\mathrm{swt}(v) : v \in \mathcal{C}^{\perp_{\mathrm{Tr}\circ\omega}} \setminus \mathcal{C}_0\};$$

(d) *(Knill–Laflamme, group form) the code corrects an error set $\mathcal{E}$ iff for all $E, F \in \mathcal{E}$ either $E^\dagger F \in \pm S$ or some element of S anti-commutes with $E^\dagger F$.*

Proof. (a) is Lemma 7.1. (b) The operators $\{E(v)\}$ for v ranging over a set of representatives of $\mathbb{F}_q^{2N}/\mathcal{C}$ are pairwise distinct and there are $q^{2N}/|\mathcal{C}|$ of them; the space decomposes into $|\mathcal{C}|$ joint eigenspaces permuted transitively, whence the dimension. (c) An error of symplectic weight w is a linear combination of Weyl operators of symplectic weight $\leq w$; such an operator is undetected iff it lies in the centralizer of S and outside S up to phases, i.e. iff its label lies in $\mathcal{C}^{\perp_{\mathrm{Tr}\circ\omega}} \setminus \mathcal{C}_0$. (d) is the standard translation of the Knill–Laflamme conditions into the group language. □

We write $[[N, K, d]]_q$ for a q-ary quantum code of length N, dimension q^K and distance d, and $[[N, K, d; c]]_q$ for an *entanglement-assisted* code consuming c maximally entangled q-ary pairs.

Theorem 7.3 (commutation must be tested over $\mathbb{F}_p$)**.** *Let $\mathcal{C}$ be $\mathbb{F}_q$-linear. Then $\mathcal{C}^{\perp_{\mathrm{Tr}\circ\omega}} = \mathcal{C}^{\perp_\omega}$, where $\perp_\omega$ is the $\mathbb{F}_q$-symplectic orthogonal. However, testing commutation only on an $\mathbb{F}_q$-basis of $\mathcal{C}$ is **invalid**: there exists a code — the $[[16,4,3]]_4$ code of Section 8 — and a weight-one operator $(0|e_0)$ which commutes with all 12 elements of an $\mathbb{F}_q$-generating set of the stabilizer yet not with all 24 elements of an $\mathbb{F}_p$-basis, because $\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(1) = 0$. The naive test reports $d = 1$ instead of $d = 3$.*

Proof. The equality of the two orthogonals is the non-degeneracy of the trace form: if $\mathrm{Tr}(\omega(x,c)) = 0$ for all $c \in \mathcal{C}$ then, by $\mathbb{F}_q$-linearity of $\mathcal{C}$ and non-degeneracy of $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}$, $\omega(x,c) = 0$ as an element of $\mathbb{F}_q$.

For the counterexample, let $\mathcal{C}$ be the stabilizer of the $[[16,4,3]]_4$ code and let $v = (0|e_0)$. Then $\omega(v,(a|b)) = -a_0$, so v commutes with $(a|b)$ iff $\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(a_0) = 0$. The set of $a_0 \in \mathbb{F}_4$ with zero absolute trace is $\{0,1\}$, a 1-dimensional $\mathbb{F}_2$-subspace of $\mathbb{F}_4$ that is *not* an $\mathbb{F}_4$-subspace. Hence commutation holds on an $\mathbb{F}_4$-generating set only if all their first-slot symbols happen to lie in $\{0,1\}$ — which is the observed degeneracy — but fails on a full $\mathbb{F}_2$-basis. The correct criterion is $\omega(v,c) = 0$ as an element of $\mathbb{F}_q$ for all $c \in \mathcal{C}$, equivalently commutation with every element of an $\mathbb{F}_p$-basis. □

7.2 Three routes

Throughout this subsection $D \subseteq \mathbb{F}_q^N$ is $\mathbb{F}_q$-linear of dimension k, $D^{\perp_E}$ is its Euclidean dual and $H_E = \mathrm{Hull}_E(D) = D \cap D^{\perp_E}$ with $h_E = \dim H_E$. We write

$$V = D \oplus D = \{(a|b) : a, b \in D\} \subseteq \mathbb{F}_q^{2N}, \qquad V^{\perp_\omega} = D^{\perp_E} \oplus D^{\perp_E}, \tag{9}$$

so that $\mathrm{rad}\, V = V \cap V^{\perp_\omega} = H_E \oplus H_E$.

Theorem 7.4 (route R1: the Galois-native twisted embedding)**.** *Let $D \subseteq \mathbb{F}_q^N$ and put $T_s(u) = (u^{[s]}J)^{\mathsf{T}}$ and*

$$\mathcal{C}_T = \{(u|T_s u) : u \in D\} \subseteq \mathbb{F}_q^{2N}.$$

Then for all $u, u' \in D$,

$$\omega\big((u|T_s u),(u'|T_s u')\big) = \langle u', u\rangle_s^{\widehat{}} - \langle u, u'\rangle_s^{\widehat{}}.$$

Consequently, if D is self-orthogonal for the transported form then $\mathcal{C}_T$ is isotropic, the operators $E(u|T_s u)$ pairwise commute, and each has order p, so that Lemma 7.2 gives a quantum code

$$[[N, N-k, d]]_q, \qquad d = \min\{\mathrm{swt}(v) : v \in \mathcal{C}_T^{\perp_{\mathrm{Tr}\circ\omega}} \setminus \mathcal{C}_T\}.$$

No coordinate transformation and no reflexivity hypothesis is used.

Proof. $(T_s u)_i = \sum_j \sigma_s(u_j) J_{ji}$, hence $u \cdot (T_s u') = \sum_{i,j} u_i \sigma_s(u'_j) J_{ji} = \langle u', u\rangle_s^{\widehat{}}$ after relabelling $i \leftrightarrow j$ and using that J is symmetric; similarly $u' \cdot (T_s u) = \langle u, u'\rangle_s^{\widehat{}}$. Subtracting gives the identity. If D is self-orthogonal both terms vanish, so $\omega = 0$ on $\mathcal{C}_T$ and Lemma 7.2(a) applies. Each operator has order p because $u \cdot (T_s u) = \langle u, u\rangle_s^{\widehat{}} = 0$. Finally $|\mathcal{C}_T| = |D| = q^k$, so $K = N - k$ by Lemma 7.2(b). □

Theorem 7.5 (the R1 distance cap)**.** *For every $b \in D^{\perp_E}$ the vector $(0|b)$ lies in $\mathcal{C}_T^{\perp_\omega}$. Consequently*

$$d_{R1} \le \min \mathrm{wt}\big(D^{\perp_E}\big).$$

Proof. $\omega((0|b),(u|T_s u)) = 0 \cdot (T_s u) - u \cdot b = -u \cdot b$, which vanishes for all $u \in D$ exactly when $b \in D^{\perp_E}$. Hence $(0|b) \in \mathcal{C}_T^{\perp_\omega}$ and $\mathrm{swt}(0|b) = \mathrm{wt}(b)$. □

Theorem 7.5 is why R1 is never the route of choice for distance. It remains the only route attached to the Galois structure when the obstruction of Theorem 6.13 applies.

Theorem 7.6 (route R2: Euclidean/CSS)**.** *Let $D' \subseteq \mathbb{F}_q^N$ be $\mathbb{F}_q$-linear of dimension k.*

(a) *If $D' \subseteq D'^{\perp_E}$ (Euclidean self-orthogonal) then $\mathcal{C} = D' \oplus D'$ is isotropic and yields $[[N, N-2k, d]]_q$ with $d = \min \mathrm{wt}(D'^{\perp_E} \setminus D')$.*

(b) *If $D'^{\perp_E} \subseteq D'$ (dual-containing) then $\mathcal{C} = D'^{\perp_E} \oplus D'^{\perp_E}$ is isotropic and yields $[[N, 2k-N, d]]_q$ with $d = \min \mathrm{wt}(D' \setminus D'^{\perp_E})$.*

Proof. Take $\mathcal{C} = A \oplus B$ with $A, B \subseteq \mathbb{F}_q^N$. Then $\omega((a|0),(0|b)) = a \cdot b$, so $\mathcal{C}$ is isotropic iff $A \subseteq B^{\perp_E}$; moreover $\mathcal{C}^{\perp_\omega} = B^{\perp_E} \oplus A^{\perp_E}$ and $\dim \mathcal{C} = \dim A + \dim B$, giving $K = N - \dim A - \dim B$ by Lemma 7.2(b). Taking $A = B = D'$ gives (a) and $A = B = D'^{\perp_E}$ gives (b).

For the distance: a vector $(a|b) \in \mathcal{C}^{\perp_\omega} \setminus \mathcal{C}$ has $a \notin A$ or $b \notin B$, and $\mathrm{swt}(a|b) \geq \max(\mathrm{wt}\, a, \mathrm{wt}\, b)$; conversely $(a|0)$ realises $\mathrm{wt}(a)$ for $a \in A^{\perp_E} \setminus A$ and $(0|b)$ realises $\mathrm{wt}(b)$ for $b \in B^{\perp_E} \setminus B$. Hence the symplectic minimum equals the stated classical minimum. □

Theorem 7.7 (route R3: entanglement-assisted)**.** *Let $D \subseteq \mathbb{F}_q^N$ be any $\mathbb{F}_q$-linear code of dimension k, and put $\mathcal{C} = V = D \oplus D$ as in (9). Then*

$$[[N, N-k-h_E,\ d\ ;\ k-h_E]]_q, \qquad d = \min \mathrm{wt}\big(D^{\perp_E} \setminus \mathrm{Hull}_E(D)\big).$$

For $h_E = k$ (i.e. D Euclidean self-orthogonal) this degenerates to $c = 0$ and reproduces Theorem 7.6(a).

Proof. The isotropic part of V is $\mathrm{rad}\, V = V \cap V^{\perp_\omega} = H_E \oplus H_E$, of dimension $2h_E$. The commutativity defect is therefore $2c = \dim V - \dim \mathrm{rad}\, V = 2k - 2h_E$, i.e. $c = k - h_E$ maximally entangled q-ary pairs.

With entanglement available, the *whole* group V becomes measurable. Decompose $\mathbb{F}_q^{2N}$ as $\mathrm{rad}\, V \oplus \bigoplus_i (\mathbb{F}_q e_i \oplus \mathbb{F}_q f_i)$ with $\{e_i, f_i\}$ hyperbolic pairs and $V = \mathrm{rad}\, V \oplus \bigoplus_i \mathbb{F}_q e_i$. For $v = s + \sum_i (\alpha_i e_i + \beta_i f_i)$ with $s \in \mathrm{rad}\, V$, the element s stabilizes the codespace outright, and each hyperbolic pair is measured with the help of one ebit: on the maximally entangled state $|\Phi\rangle$ one has $(A \otimes I)|\Phi\rangle = (I \otimes A^{\mathsf{T}})|\Phi\rangle$, so $E(e_i) \otimes I$ and $I \otimes E(f_i)^{\mathsf{T}}$ act identically and the pair is simultaneously measurable. Hence $v \otimes I$ fixes the codespace.

Consequently an error F is undetected exactly when it commutes with all of V **and** lies outside V — note: outside V, not merely outside $\mathrm{rad}\, V$ — i.e. when its label lies in $V^{\perp_\omega} \setminus V$. By (9) and the argument of Theorem 7.6, the symplectic minimum over $V^{\perp_\omega} \setminus V$ equals $\min \mathrm{wt}(D^{\perp_E} \setminus \mathrm{Hull}_E(D))$. The dimension is $K = N - \dim V + c = N - 2k + (k - h_E) = N - k - h_E$.

For $h_E = k$ we get $c = 0$, $K = N - 2k$ and $D^{\perp_E} \setminus \mathrm{Hull}_E(D) = D^{\perp_E} \setminus D$, which is Theorem 7.6(a). □

Remark 7.8 (field discipline). $\mathcal{C} = D \oplus D$ is $\mathbb{F}_q$-linear, so by Theorem 7.3 the trace-symplectic and the $\mathbb{F}_q$-symplectic orthogonals coincide and the entanglement count $c = k - h_E$ is an $\mathbb{F}_q$-quantity. For the *semilinear* group $\mathcal{C}_T$ of R1 the standard q-ary entanglement formula does not apply unchanged, because $\mathcal{C}_T$ is only $\mathbb{F}_p$-linear; we do not use it.

7.3 The parameter theorem

Theorem 7.9 (parameters)**.** *Let $C \leq R_n$, $D = \varphi_n(C)$, $N = 4tn$, $k = \dim_{\mathbb{F}_q} D$ and $h_E = \dim(D \cap D^{\perp_E})$. Then:*

(a) ***(R1)*** *if $D \subseteq D^{\widehat{\perp_s}}$ then $[[N, N-k, d]]_q$ with $d \leq \min \mathrm{wt}(D^{\perp_E})$;*

(b) ***(R2)*** *if D' is Euclidean self-orthogonal of dimension k then $[[N, N-2k, d]]_q$, and if D' is dual-containing then $[[N, 2k-N, d]]_q$; here $D' = D$ when $t = 1$ and $r \mid s$, and $D' = \Psi(D)$ when a congruence as in Proposition 6.12 exists;*

(c) ***(R3)*** *for arbitrary D, $[[N, N-k-h_E, d; k-h_E]]_q$.*

In all cases, writing G for the group ($\mathcal{C}_T$ for R1, V for R2/R3),

$$d = \min\{\mathrm{swt}(v) : v \in G^{\perp_\omega} \setminus G\}.$$

All dimensions are over $\mathbb{F}_q$, the length is $N = 4tn$, and the alphabet is $\mathbb{F}_q$.

Proof. Combine Theorem 6.15 for the length, Theorems 7.4, 7.6 and 7.7 for the three routes, and Lemma 7.2(c) for the distance formula. Note that the reduction of the symplectic minimum to a classical minimum is proved for R2 and R3 only; for R1 only the upper bound of Theorem 7.5 is available, and we state it as such. □

Remark 7.10 (which route to use). R2 is the primary route whenever the transported form is already Euclidean or a congruence exists: it is the only route whose distance is not capped by $\min \mathrm{wt}(D^{\perp_E})$, and it consumes no entanglement. R1 is the Galois-native route for $t \geq 2$ when no congruence exists — precisely the case of the non-principal hull witnesses — accepting the cap of Theorem 7.5. R3 converts *any* image code, including an LCD one ($h_E = 0$), into a quantum code, at the price of entanglement.

7.4 Worked examples V: the $\mathbb{F}_p$-basis subtlety

Example 7.11 (why commutation must be tested over $\mathbb{F}_p$)**.** Let $q = 4$, $N = 16$, and consider the $[[16, 4, 3]]_4$ code produced at $(q, t, n) = (4, 1, 4)$ by the self-orthogonal route: the image code $D \subseteq \mathbb{F}_4^{16}$ has $k = 6$, $h_E = 6$, so $K = 16 - 2 \cdot 6 = 4$. The stabilizer group is $\mathcal{C} = D \oplus D$, of $\mathbb{F}_4$-dimension 12.

Take the weight-one operator $E(0|e_0)$ labelled by $v = (0|e_0)$, where $e_0 = (1, 0, \ldots, 0)$. For $(a|b) \in \mathcal{C}$,

$$\omega\big(v, (a|b)\big) = 0 \cdot b - a \cdot e_0 = -a_0.$$

So $E(v)$ commutes with $E(a|b)$ iff $\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(a_0) = 0$. Now

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(1) = 1 + 1^2 = 0, \qquad \mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega) = \omega + \omega^2 = 1,$$

so the set of $a_0 \in \mathbb{F}_4$ with vanishing trace is $\{0, 1\}$: a 1-dimensional $\mathbb{F}_2$-subspace of $\mathbb{F}_4$ that is *not* an $\mathbb{F}_4$-subspace. Consequently:

- testing commutation on the 12 elements of an $\mathbb{F}_4$-generating set of $\mathcal{C}$ reports that v commutes with everything, and the naive distance comes out as $d = 1$;
- testing it on the 24 elements of an $\mathbb{F}_2$-basis detects that some $(a|b)$ has $a_0 = \omega$, so v does *not* commute, and the correct distance $d = 3$ is obtained.

By Theorem 7.3 the two orthogonals $\mathcal{C}^{\perp_{\mathrm{Tr}\circ\omega}}$ and $\mathcal{C}^{\perp_\omega}$ *coincide* for an $\mathbb{F}_q$-linear $\mathcal{C}$; what fails is not the orthogonal but the *test*. Every distance reported in this paper is computed with the $\mathbb{F}_p$-basis test.

Example 7.12 (the three routes on one family)**.** Take $(q, t, n) = (9, 1, 6)$, so $N = 24$. The search produces, among others, two image codes $D \subseteq \mathbb{F}_9^{24}$ with very different hulls.

First, $k = \dim D = 8$ and $h_E = 8$, i.e. $D \subseteq D^{\perp_E}$: the code is Euclidean self-orthogonal. Route R2(a) (Theorem 7.6) applies with $A = B = D$ and gives

$$K = N - 2k = 24 - 16 = 8, \qquad d = \min \mathrm{wt}\big(D^{\perp_E} \setminus D\big) = 4, \qquad [[24, 8, 4]]_9.$$

Feeding the same D to route R3 instead gives $c = k - h_E = 0$ and $K = N - k - h_E = 8$: the same code, which is the degeneracy stated at the end of Theorem 7.7. Note $D^{\perp_E} \setminus D = D^{\perp_E} \setminus \mathrm{Hull}_E(D)$ here, since $\mathrm{Hull}_E(D) = D$; so the coset-distance theorem (Theorem 8.1) is the same statement for R2 and R3 in this case.

Second, $k = 20$ and $h_E = 2$, so D is neither self-orthogonal nor dual-containing and R2 does not apply at all. Route R3 gives

$$c = k - h_E = 18, \qquad K = N - k - h_E = 24 - 20 - 2 = 2, \qquad [[24, 2, 10; 18]]_9.$$

No code found anywhere in the search has $d > 10$; this $[[24, 2, 10; 18]]_9$ and the $[[20, 2, 10; 18]]_4$ certified in Example 8.12 are the two largest-distance codes of the search, each costing 18 ebits.

The two rows are the two ends of the trade-off made explicit by Theorem 8.5: with $d \geq 4$ and $K \geq 2$ one needs $2k \leq N + c_0 - K_0$, so a large k (which is what makes $D^{\perp_E}$ small and hence d large) forces a large entanglement budget c_0. The stabilizer code pays nothing and reaches $d = 4$; the entanglement-assisted code reaches $d = 10$ and pays $c = 18$.

8 Distance certification and quantum parameter analysis

8.1 The coset-distance theorem

The distance formulas of Theorems 7.6 and 7.7 look like minima over a set difference. The following theorem says that they are something better: the minimum weight of an ordinary linear code.

Theorem 8.1 (coset-distance characterisation)**.** *Let $D \subseteq \mathbb{F}_q^N$ be $\mathbb{F}_q$-linear, $H_E = D \cap D^{\perp_E}$ and $K = N - k - h_E$. Then, for routes R2 and R3,*

$$d = \min\bigl\{\mathrm{wt}(x) : x \in D^{\perp_E} \setminus \mathrm{Hull}_E(D)\bigr\}$$

is the minimum weight of the quotient code

$$D^{\perp_E} / \mathrm{Hull}_E(D),$$

a linear code of length N and dimension $\dim D^{\perp_E} - \dim \mathrm{Hull}_E(D) = (N - k) - h_E = K$. Equivalently, d is the minimum weight of the set of nonzero cosets of $\mathrm{Hull}_E(D)$ in $D^{\perp_E}$. In particular, for any integer $\delta \geq 1$,

$$d \geq \delta \iff \textit{every } x \in D^{\perp_E} \textit{ with } \mathrm{wt}(x) < \delta \textit{ lies in } \mathrm{Hull}_E(D). \tag{10}$$

Proof. By Theorem 7.9, for R2 and R3 d is the minimum of swt over $G^{\perp_\omega} \setminus G$ with $G = V$. Since $\mathrm{swt}(a|b) \geq \max(\mathrm{wt}\, a, \mathrm{wt}\, b)$ and since $(a|0) \in V^{\perp_\omega} \setminus V$ whenever $a \in D^{\perp_E} \setminus D$, that symplectic minimum equals $\min\{\mathrm{wt}(x) : x \in D^{\perp_E} \setminus D\}$; and for $x \in D^{\perp_E}$ one has $x \in D$ iff $x \in D \cap D^{\perp_E} = H_E$, so $D^{\perp_E} \setminus D = D^{\perp_E} \setminus H_E$ as sets.

Now, the quotient group $D^{\perp_E}/H_E$ is an $\mathbb{F}_q$-vector space of dimension $\dim D^{\perp_E} - \dim H_E = (N-k) - h_E = K$, and its nonzero elements are exactly the cosets $x + H_E$ with $x \in D^{\perp_E} \setminus H_E$. The weight of a coset is well defined as $\mathrm{wt}(x+H_E) = \min\{\mathrm{wt}(y) : y \in x+H_E\}$, and its minimum over nonzero cosets is exactly $\min\{\mathrm{wt}(x) : x \in D^{\perp_E} \setminus H_E\}$: indeed $\mathrm{wt}(x + H_E) \leq \mathrm{wt}(x)$ gives one inequality, and conversely any $y \in x + H_E$ with $y \neq 0$ may be substituted for x. This is the assertion. The criterion (10) is the contrapositive restatement. □

Remark 8.2. For route R1 only the upper bound of Theorem 7.5 is available, and Theorem 8.1 is *not* claimed for R1: the group $\mathcal{C}_T$ is semilinear, the set difference is not a quotient by an $\mathbb{F}_q$-subspace of the ambient space in the same way, and we do not use it.

8.2 The coset Singleton bound

Theorem 8.3 (coset Singleton bound)**.** *Let $D \subseteq \mathbb{F}_q^N$ be $\mathbb{F}_q$-linear and let $K = N - k - h_E$ be the dimension of the quantum code produced by route R2 or R3. Then*

$$K + d \leq N + 1. \tag{11}$$

Proof. Consider the quotient code $\mathcal{Q} = D^{\perp_E}/\operatorname{Hull}_E(D)$ of Theorem 8.1: it has length N, dimension K and minimum weight d. Let $S \subseteq \{1, \ldots, N\}$ be any set of $N - d + 1$ coordinates and let π_S be the projection onto those coordinates. If π_S were not injective on $\mathcal{Q}$, there would be a nonzero coset $x + H_E$ with $\pi_S(x + H_E) = 0$, i.e. an element $y \in x + H_E$ supported outside S, i.e. supported on the remaining $d - 1$ coordinates. Then $\operatorname{wt}(y) \leq d - 1 < d$, contradicting the definition of d as the minimum weight of a nonzero element of $\mathcal{Q}$. Hence π_S is injective, so $q^K = |\mathcal{Q}| \leq q^{N-d+1}$ and therefore $K \leq N - d + 1$. □

Proposition 8.4 (the coset Singleton bound is new)**.** *The bound (11) is not the entanglement-assisted Singleton bound, and neither implies the other.*

Proof. The entanglement-assisted Singleton bound reads $N - K + c \geq 2(d - 1)$, i.e. $K \leq N + c - 2d + 2$. For the code $[[20, 2, 10; 18]]_4$ constructed in Section 9 one has $N = 20$, $K = 2$, $d = 10$, $c = 18$: the coset Singleton bound gives $2 \leq 20 - 10 + 1 = 11$, which holds, while the EA Singleton bound gives $2 \leq 20 + 18 - 20 + 2 = 20$, which is much weaker. Conversely, for a code with small c and small d, say $[[24, 20, 2; 0]]_4$ ($N = 24$, $K = 20$, $d = 2$, $c = 0$), the coset Singleton bound gives $20 \leq 24 - 2 + 1 = 23$, while the EA Singleton bound gives $20 \leq 24 + 0 - 4 + 2 = 22$ — the EA bound is now the stronger one. Hence neither implies the other. □

8.3 The feasibility inequality

Theorem 8.5 (feasibility inequality)**.** *Suppose a code produced by route R3 from $D \subseteq \mathbb{F}_q^N$ has $d \geq 4$, $c \leq c_0$ and $K \geq K_0$. Then*

$$2k \leq N + c_0 - K_0. \tag{12}$$

Proof. By Theorem 7.7, $c = k - h_E$ and $K = N - k - h_E$. The hypotheses $c \leq c_0$ and $K \geq K_0$ read $h_E \geq k - c_0$ and $h_E \leq N - k - K_0$. Hence $k - c_0 \leq N - k - K_0$, i.e. $2k \leq N + c_0 - K_0$. □

Remark 8.6 (what the inequality does and does not say). The hypothesis $d \geq 4$ enters only through the *motivation* for imposing the constraints, not through the algebra: the implication "$c \leq c_0$ and $K \geq K_0 \Rightarrow$ (12)" holds for every code, whatever its distance. Inequality (12) is one-way: it is a necessary condition, not a sufficient one, and no converse is claimed.

Its content is the trade-off behind the entanglement budget: cheap entanglement means a large Euclidean hull, and a large Euclidean hull means few logical qudits. Combined with the criterion (10), it converts the question "is there a family with $d \geq 4$, bounded c and $K \geq 2$?" into two explicit conditions on D: (i) k must be close to $N/2$, and (ii) every vector of weight at most 3 in $D^{\perp_E}$ must lie in $\operatorname{Hull}_E(D)$.

8.4 Bookkeeping identities, and an erratum

Theorem 8.7 (bookkeeping)**.** *For every code produced by any of the three routes from $C \leq R_n$ with $D = \varphi_n(C)$:*

(a) *$k = \dim_{\mathbb{F}_q} D$ is even, and $N = 4tn$ is divisible by 4;*

(b) *$K - c = N - 2k \equiv 0 \pmod 4$; in particular $K \equiv c \pmod 4$;*

(c) $c + K = N - 2h_E$;

(d) $c \geq 2k - N$.

Proof. (a) Under the Morita correspondence (2), D is the image of I_W with $W \leq B^2$, and $\dim_{\mathbb{F}_q} I_W = 2\dim_{\mathbb{F}_q} W$; hence k is even. (b) $K - c = (N - k - h_E) - (k - h_E) = N - 2k$, which is divisible by 4 by (a) and $N = 4tn$. (c) is immediate from the two defining identities. (d) From $c = k - h_E$ and $h_E \leq \min(k, N - k)$ (as h_E is the dimension of a subspace of both D and $D^{\perp_E}$) we get $c \geq k - (N - k) = 2k - N$. □

Remark 8.8 (erratum). An earlier version of this project stated the identity as $c + K = N - k$. That is wrong: it would force $h_E = k/2$, which is false in general, and it is contradicted by 23 of the 25 rows of the earlier paper's own parameter table. The correct identity is Theorem 8.7(c). We record this as an erratum rather than repair it silently; no *parameter* computed in the earlier work is affected, because K and c were always computed from k and h_E directly.

8.5 Certification methodology

Every distance reported in this paper is either **exact** or a **proved lower bound**, and the two are never conflated. Three independent methods are used, and every headline code is certified by all three.

M1 — complete enumeration. Enumerate the relevant classical difference set (the coset code of Theorem 8.1, or the classical difference set of R2) completely and take the minimum weight. This is exact and complete whenever $q^{\dim} \leq 3 \cdot 10^6$.

M2 — increasing-weight support search. A different algorithm: for each support of size $w = 1, 2, 3, \ldots$ in increasing order, solve the left-kernel problem asking whether the difference set contains a vector supported inside it. The first w at which a solution exists is the distance. Because the algorithm differs structurally from M1, agreement of the two is a genuine cross-check.

M3 — ambient quantum search. Search the ambient space $\mathbb{F}_q^{2N}$ in increasing symplectic weight for a vector v with $v \in V^{\perp_\omega}$ and $v \notin V$. This certifies the distance directly in the quantum language, without passing through Theorem 8.1. It is complete only up to the weight reached, and is reported as a lower bound otherwise.

In addition, for every certified code: the stabilizer group is verified abelian on an $\mathbb{F}_p$-basis (Theorem 7.3); K and c are recomputed from k and h_E; and — crucially — *the certified code is the exact code found by the search*, the image basis being carried through the whole pipeline. An earlier version of this project certified a re-selected code with the same invariants instead of the reported one; that defect was found and fixed, and the current pipeline cannot reproduce it.

8.6 Parameter analysis

The computation covers $q \in \{4, 8, 9\}$ with $tn \leq 8$ (42 families), of which 23 are enumerated exhaustively, 4 are counted but not enumerated, and 15 are blocked by the component enumeration. Ten families that earlier work could only sample are now exhaustive. In total 120 721 codes have computed invariants and 5 055 have a computed distance, of which 4 418 are exact and 637 are proved lower bounds. Every one of the 5 055 satisfies the entanglement-assisted Singleton and Hamming bounds, and the coset Singleton bound (11) (0 violations in each case).

Table 2: Headline codes. S-def is the entanglement-assisted Singleton defect $N-K+c-2(d-1)$. All distances exact except the last row of the upper block, which is a proved lower bound.

code	family (q,t,n)	route	k	h_E	K	d	c	S-def
$[[24,8,4]]_9$	$(9,1,6)$	R2-SO	8	8	8	4	0	10
$[[20,4,4]]_4$	$(4,1,5)$	R2-SO	8	8	4	4	0	10
$[[20,4,4]]_4$	$(4,1,5)$	R2-DC	12	8	4	4	0	10
$[[16,4,3]]_4$	$(4,1,4)$	R2-SO	6	6	4	3	0	8
$[[16,4,3]]_4$	$(4,1,4)$	R2-DC	10	6	4	3	0	8
$[[20,2,10;18]]_4$	$(4,1,5)$	R3	18	0	2	10	18	18
$[[24,2,10;18]]_9$	$(9,1,6)$	R3	20	2	2	10	18	22
$[[20,4,9;16]]_9$	$(9,1,5)$	R3	16	0	4	9	16	16
$[[24,4,9;20]]_4$	$(4,1,6)$	R3	20	0	4	9	20	24
$[[16,2,8;14]]_9$	$(9,1,4)$	R3	14	0	2	8	14	14
$[[16,2,7;10]]_9$	$(9,1,4)$	R3	12	2	2	7	10	12
$[[12,2,6;10]]_4$	$(4,1,3)$	R3	10	0	2	6	10	10
$[[24,4,6;20]]_4$	$(4,2,3)$	R3	20	0	4	6	20	30
$[[16,4,5;8]]_4$	$(4,1,4)$	R3	10	2	4	5	8	12
$[[24,10,4;2]]_9$	$(9,1,6)$	R3	8	6	10	≥ 4	2	10
$[[24,6,2;2]]_9$	$(9,2,3)$	R3	10	8	6	2	2	18
$[[24,14,2]]_9$	$(9,2,3)$	R1	10	—	14	2	0	8
$[[32,8,2;4]]_4$	$(4,2,4)$	R3	14	10	8	2	4	26

Proposition 8.9 (frontier, computational)**.** *Among the* 5 055 *codes with a distance there are* 368 *distinct tuples* (N,K,d,c)*;* 88 *of them are non-dominated and* 280 *are dominated, where* (N,K,d,c) *dominates* (N',K',d',c') *iff* $N \leq N'$*,* $K \geq K'$*,* $d \geq d'$*,* $c \leq c'$ *with at least one strict inequality, the comparison being made inside one alphabet.*

Remark 8.10 (what the frontier is and is not). Proposition 8.9 is a statement about the codes *scanned*, not about all codes obtainable from the construction: the families that were counted but not enumerated, and the fifteen that were blocked, are excluded, and they are listed in the supplementary material. No tuple outside the scanned families is claimed to be dominated.

Table 2 lists the headline codes, all certified by the three methods. Table 5 in Appendix A lists the full certification table.

The last three rows are the non-principal-hull witnesses of Theorem 5.4: their hulls have $\mu_B = 3$. They are included for their structural significance, not for their parameters ($d = 2$). The middle one of the three is a *stabilizer* code produced by route R1, which is the only route available there because Theorem 6.13 blocks the Euclidean one at $(q,t) = (9,2)$.

8.7 Bounds satisfied

Proposition 8.11 (bounds)**.** *All* 5 055 *codes with a computed distance satisfy*

(a) *the entanglement-assisted Singleton bound* $N - K + c \geq 2(d-1)$*;*

(b) *the entanglement-assisted Hamming bound with effective dimension* $K-c$*:* $\big(\sum_{i=0}^{t} \binom{N}{i}(q^2-1)^i\big) q^{K-c} \leq q^N$ *with* $t = \lfloor (d-1)/2 \rfloor$*;*

(c) *the coset Singleton bound* $K + d \leq N + 1$*.*

A lower-bound distance is tested with its proved value.

Table 3: Minimum entanglement c for each distance, over the whole search, with the attaining code. The distance range is $d = 2, \ldots, 8$: these are the distances for which Part 4(c) of the Phase-7 log records a minimum-entanglement attaining code. The two larger distances of the search, $d = 9$ and $d = 10$, are attained only at $c = 16$ and $c = 18$ (Table 2).

d	minimal c	attaining code	status of d
2	0	$[[24, 20, 2; 0]]_4$	exact
3	0	$[[24, 12, 3; 0]]_9$	exact
4	0	$[[24, 8, 4; 0]]_9$	exact
5	2	$[[24, 6, 5; 2]]_9$	proved lower bound
6	10	$[[12, 2, 6; 10]]_4$	exact
7	10	$[[16, 2, 7; 10]]_9$	exact
8	14	$[[16, 2, 8; 14]]_9$	exact

8.8 The entanglement budget

Table 3 is the measured answer to the question behind the criterion of Theorem 8.5: what is the least entanglement at which a given distance becomes available.

Reading Table 3 with Theorem 8.1: $d = 4$ is available with *no* entanglement at all ($[[24, 8, 4]]_9$, a stabilizer code), while $d \geq 5$ costs entanglement, and the cost grows steeply. With the entanglement bounded by $c \leq 4$ the best attainable distance is $d = 4$, attained with $K = 10$ and $c = 2$ by $[[24, 10, 4; 2]]_9$ — a proved lower bound, and the frontier point for the criterion.

8.9 Worked example VI: a full certification

Example 8.12 (certifying $[[20, 2, 10; 18]]_4$)**.** Take the family $(q, t, n) = (4, 1, 5)$, so $N = 4 \cdot 1 \cdot 5 = 20$. The search finds a cyclic code $C \leq R_n$ whose Gray image $D = \varphi_5(C) \subseteq \mathbb{F}_4^{20}$ has

$$k = \dim_{\mathbb{F}_4} D = 18, \qquad h_E = \dim\bigl(D \cap D^{\perp_E}\bigr) = 0.$$

Route R3 (Theorem 7.7) then gives

$$c = k - h_E = 18, \qquad K = N - k - h_E = 20 - 18 - 0 = 2,$$
$$[[20, 2, d; 18]]_4.$$

The distance. By Theorem 8.1, d is the minimum weight of the quotient code $\mathcal{Q} = D^{\perp_E} / \operatorname{Hull}_E(D)$; here $h_E = 0$, so $\operatorname{Hull}_E(D) = \{0\}$ and $\mathcal{Q} = D^{\perp_E}$, a linear $[20, 2]_4$ code. Its $4^2 = 16$ codewords are enumerated completely (method M1), and the minimum weight found is 10; the independent support search (M2) and the ambient quantum search (M3, which certifies $d \geq 3$ as a lower bound) agree with the reported value. Hence $d = 10$ exactly and

$$[[20, 2, 10; 18]]_4.$$

The three bound checks:

(i) entanglement-assisted Singleton: $N - K + c = 20 - 2 + 18 = 36 \geq 2(d - 1) = 18$, with defect 18;

(ii) entanglement-assisted Hamming with effective dimension $K - c = -16$: $\bigl(\sum_{i=0}^{4} \binom{20}{i} (4^2 - 1)^i\bigr) 4^{-16} \leq 4^{20}$;

(iii) coset Singleton (Theorem 8.3): $K + d = 2 + 10 = 12 \leq N + 1 = 21$.

For comparison, the entanglement-assisted Singleton bound alone only gives $K \le N + c - 2d + 2 = 20$, so the coset Singleton bound is the binding one here (Proposition 8.4).

Finally, the bookkeeping identities of Theorem 8.7: $k = 18$ is even; $K - c = 2 - 18 = -16 = N - 2k = 20 - 36$ is divisible by 4; and $c + K = 20 = N - 2h_E$, in accordance with the corrected identity — the earlier formula $c + K = N - k$ would give $20 = 2$, which is false.

No code found anywhere in the search has $d > 10$; together with the $[[24, 2, 10; 18]]_9$ of Example 7.12, this $[[20, 2, 10; 18]]_4$ is one of the two largest-distance codes of the search, and it is also the most entanglement-hungry ($c = 18$ out of $N = 20$). That trade-off is the content of Theorem 8.5: $d \ge 4$ together with $K \ge 2$ forces $2k \le N + c_0 - K_0$, i.e. large d and moderate K cannot both be had at small c.

9 Comparison with existing quantum codes

9.1 The terms of the comparison

The comparison is with the tables of [5], retrieved 2026-11-04 (page last changed 10.06.2024). Two limitations must be stated before any number is quoted, and we state them where they bite.

(1) **The tables cover stabilizer codes only.** There is no entanglement-assisted table: the corresponding endpoint returns HTTP 404. Every entanglement-assisted code of this paper is therefore *unbenchmarked*. This is a substantial blind spot, since the entanglement-assisted route is where the largest distances of this paper occur.

(2) **Only $q = 4$ and $q = 9$ are tabulated.** For $q = 8$ the site returns "currently not available for q=8". Hence the $(8, \cdot, \cdot)$ codes of this paper are unbenchmarked as well. (A third point of convention: the site prints the *prime* p in its code labels, so that $q = 4$ appears as $[[N, K]]_2$ and $q = 9$ as $[[N, K]]_3$, while the construction block says "over GF(p^2)". An earlier phase of this project misread this and concluded, incorrectly, that $q = 9$ was not covered; we record that as an erratum in the supplementary material and the corrected comparison is the one below.)

9.2 The comparison table

9.3 Reading of the comparison

Every stabilizer code constructed here is at best *equal* to the best known distance, and then only with $d = 2$, where equality is uninformative. Every code with $d \ge 3$ is one or two below the best known value. **No record parameter is claimed**, and none is implied: the codes are presented for their origin, not for their position in the tables.

That said, the comparison does say something. The framework produces, at $N = 24$ over $\mathbb{F}_9$, a stabilizer code of distance 4 where the best known is 6 (and where the upper bound is 8): the construction is within a factor of 1.5 of the best known code at that length and dimension, and it does so from a chain of entirely algebraic operations over a non-commutative ring. More importantly, the *entanglement-assisted* codes — where this framework's distances are largest ($d = 10$, attained at both $[[20, 2, 10; 18]]_4$ and $[[24, 2, 10; 18]]_9$) — cannot be compared at all, and the absence of an entanglement-assisted table is a limitation of the literature rather than of the construction.

9.4 Novelty

We do not claim that matrix rings, Galois hulls, Gray maps, quasi-cyclic images, stabilizer codes or entanglement-assisted codes are new; each is attributed in Section 2. The novelty

Table 4: Every stabilizer code ($c = 0$) produced by the framework, compared with [5]. "Best known" is the lower .. upper bound pair for $[[N, K]]_q$.

code	family (q, t, n)	route	best known $[[N, K]]_q$ / gap
$[[16, 12, 2]]_4$	$(4, 1, 4)$	R2-SO	2..2 — meets
$[[16, 12, 2]]_4$	$(4, 1, 4)$	R2-DC	2..2 — meets
$[[20, 16, 2]]_4$	$(4, 1, 5)$	R2-SO	2..2 — meets
$[[20, 16, 2]]_4$	$(4, 1, 5)$	R2-DC	2..2 — meets
$[[24, 20, 2]]_4$	$(4, 1, 6)$	R2-DC	2..2 — meets
$[[20, 12, 2]]_4$	$(4, 1, 5)$	R2-SO	3..3 — 1 below
$[[20, 12, 2]]_4$	$(4, 1, 5)$	R2-DC	3..3 — 1 below
$[[24, 16, 2]]_4$	$(4, 1, 6)$	R2-DC	3..3 — 1 below
$[[16, 4, 3]]_4$	$(4, 1, 4)$	R2-SO	5..5 — 2 below
$[[16, 4, 3]]_4$	$(4, 1, 4)$	R2-DC	5..5 — 2 below
$[[20, 4, 4]]_4$	$(4, 1, 5)$	R2-SO	6..6 — 2 below
$[[20, 4, 4]]_4$	$(4, 1, 5)$	R2-DC	6..6 — 2 below
$[[16, 8, 3]]_9$	$(9, 1, 4)$	R2-DC	4..4 — 1 below
$[[24, 8, 4]]_9$	$(9, 1, 6)$	R2-SO	6..8 — 2 below (best known 6)
$[[24, 12, 3]]_9$	$(9, 1, 6)$	R2-DC	5..6 — 2 below (best known 5)

claim is the following, specific and proved.

(1) **The duality.** Over the non-commutative ring $R = M_2(A_t)$, the Galois twist must be $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$, an *anti*-automorphism. Over a commutative ring this degenerates to σ_s and there is nothing to prove; here it is what makes the dual of a *right* module a right module with the correct size (Theorems 4.3 and 3.4).

(2) **Hulls without principality, and the refutation.** The entire hull theory is developed for arbitrary right ideals, and the natural conjecture that Galois hulls are principal is **false** (Theorem 5.4), with the phenomenon confined by the structural condition $t \geq 2$ and $p \mid n$. This has no commutative analogue.

(3) **The exact induced form.** The coefficient expansion transports the Galois pairing to $u^{[s]}(I_{4n} \otimes H_t)v^{\mathsf{T}}$ (Theorem 6.3) — a $\widehat{\sigma}_s$-Hermitian form with a Hankel Gram matrix, not the Euclidean or plain Hermitian form unless $t = 1$. Duality and hulls are preserved including on the non-principal hulls (Theorems 6.5 and 6.6).

(4) **Self-orthogonality without reflexivity.** Theorem 6.10 is proved by a one-sided argument, so the whole construction is available in the general Galois regime.

(5) **Two new bounds.** The coset-distance characterisation (Theorem 8.1) and the coset Singleton bound (Theorem 8.3), the latter being neither the entanglement-assisted Singleton bound nor implied by it (Proposition 8.4).

(6) **Quantum codes from non-principal hulls.** The codes $[[24, 6, 2; 2]]_9$ and $[[32, 8, 2; 4]]_4$ of Table 2 come from hulls with $\mu_B = 3$. Their parameters are weak; their existence is the point, since it shows that no principality hypothesis is hidden anywhere in the chain.

Relative to the closest prior work [13] — Galois hulls, a Gray map, and quantum codes over a finite non-chain ring — the differences are: their ring is *commutative*, so no anti-automorphism arises and no principality question can be asked; their Gray image has length tn, whereas ours has length $4tn$ (Theorem 6.15), because $\dim_{\mathbb{F}_q} M_2(A_t) = 4t$; they determine hull *dimensions*,

whereas we construct codes from hulls that are not principal and classify when $\mu_B \geq 3$ can occur; and we prove the two bounds of item (5).

9.5 On the earlier tuple $[[24, 4, 5; 12]]_4$

An earlier phase of this project reported $[[24, 4, 5; 12]]_4$, obtained at $(q, t, n) = (4, 2, 3)$ from a 300-code sample. Exhaustive enumeration of that family (35 937 codes) shows that its best entanglement-assisted distance is 6, attained by $[[24, 4, 6; 20]]_4$ ($k = 20$, $h_E = 0$, $c = 20$, $K = 4$): the earlier value was a sampled lower bound, not an optimum. The tuple $[[24, 4, 5; 12]]_4$ is not reproduced by the exhaustive search and is dominated (by $[[20, 4, 6; 12]]_4$). We report this as a change of value obtained by a stronger method, not as a retraction: nothing in the earlier derivation was wrong, it was simply incomplete.

10 Conclusion

We have given a complete and self-contained treatment of the chain

$$\text{non-commutative matrix rings} \longrightarrow \text{cyclic codes} \longrightarrow \text{Galois duality and hulls}$$
$$\longrightarrow \text{Gray maps} \longrightarrow \text{quantum codes.}$$

The algebraic core is that over $R = M_2(A_t)$ the Galois twist must be the anti-automorphism $\theta_s = \sigma_s \circ (\cdot)^{\mathsf{T}}$, so that the dual of a right ideal is a right ideal with the correct dimension (Theorems 4.3, 4.6, 3.4). The hull theory is developed for arbitrary — in particular non-principal — right ideals, and the natural conjecture that Galois hulls are principal is false (Theorem 5.4), with the phenomenon confined to $t \geq 2$ and $p \mid n$.

The coefficient expansion transports the Galois pairing to the form

$$u^{[s]}(I_{4n} \otimes H_t)v^{\mathsf{T}}$$

(Theorem 6.3); duality and hulls are transported exactly, non-principal hulls included; and the image of a Galois hull is self-orthogonal *without* any reflexivity hypothesis (Theorem 6.10), which is what makes the whole chain work in the general Galois regime. A structural obstruction rules out the Euclidean coordinate change in characteristic two with t even (Theorem 6.13), and its hypotheses are exact.

On the quantum side, three routes (R1–R3) are available, their parameters and distances are given with the hypotheses visible, and the distance is characterised as the minimum weight of a quotient code (Theorem 8.1), yielding the coset Singleton bound $K + d \leq N + 1$ (Theorem 8.3). A feasibility inequality (Theorem 8.5) describes the entanglement budget, and the measured frontier is reported (Proposition 8.9, Table 3).

The best codes obtained are $[[24, 8, 4]]_9$ among stabilizer codes and $[[20, 2, 10; 18]]_4$ and $[[24, 2, 10; 18]]_9$ among entanglement-assisted codes. **No record parameter is claimed.** The stabilizer codes are one or two below the best known distances of [5], and the entanglement-assisted codes cannot be compared at all, since no entanglement-assisted table exists. What is claimed is a new *origin* for quantum codes: non-commutative Galois hulls, transported exactly, with non-principal hulls admitted and with two new bounds on the resulting parameters.

Open problems. (i) Determine the Galois hull *dimension* distribution in the wild families, where the cyclic-code enumeration has resisted complete treatment for 15 families. (ii) Find a closed or recursive description of the set of weights of the quotient code of Theorem 8.1 in terms of the CRT data of C. (iii) Prove or refute the two conjectures stated at the end of Section 5: $\mathrm{Hull}_s(C) \subseteq \sigma_{2s}(C)$, and $\dim \mathrm{Hull}_s(C) = \dim \mathrm{Hull}_s(C^{\perp_s})$. (vi) Decide the sharper form (O-3.1b) of the refuted conjecture (Remark 5.6): every witness currently known is a code

that is itself non-principal, so the question whether a *principal* cyclic code can have a non-principal Galois hull in the wild regime is open. (iv) Determine whether the obstruction of Theorem 6.13 is the only one: we have not proved that a congruence exists for all t odd in characteristic two, only that it exists for $t = 1, 3, 5$ at $q = 4, 8$. (v) Extend the comparison to $q = 8$ and to entanglement-assisted codes, neither of which is tabulated.

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A Certification table and comparison data

A.1 The certification table

Table 5 is the complete output of the certification stage: all 28 certified codes, reproduced verbatim from the computation log (`04_COMPUTATION/phase7_extended_search_log.txt`, Part 6). The columns M1, M2, M3 record the three independent distance methods of Section 8: a dash means the method could not complete, and "> 2" means the ambient quantum search (M3) reached weight 2 without finding a vector, i.e. it certifies $d \geq 3$ as a *lower* bound only. The column "exact" says yes when all three methods that completed agree with the reported value, the stabilizer is $\mathbb{F}_p$-abelian, and K and c recompute from k and h_E.

Table 5: All 28 certified codes (Part 6 of the Phase-7 log, verbatim). S-def is the entanglement-assisted Singleton defect $N - K + c - 2(d-1)$.

code	family	route	k	K	d	c	S-def	M1 / M2 / M3
$[[24, 8, 4]]_9$	$(9, 1, 6)$	SO	8	8	4	0	10	– / 4 / > 2
$[[20, 4, 4]]_4$	$(4, 1, 5)$	SO	8	4	4	0	10	– / 4 / > 2
$[[20, 4, 4]]_4$	$(4, 1, 5)$	DC	12	4	4	0	10	– / 4 / > 2
$[[20, 2, 10; 18]]_4$	$(4, 1, 5)$	EA	18	2	10	18	18	10 / n/a / > 2
$[[24, 2, 10; 18]]_9$	$(9, 1, 6)$	EA	20	2	10	18	22	10 / n/a / > 2
$[[4, 2, 2; 2]]_4$	$(4, 1, 1)$	EA	2	2	2	2	2	2 / n/a / 2
$[[8, 4, 3; 4]]_4$	$(4, 1, 2)$	EA	4	4	3	4	4	3 / n/a / 3
$[[12, 2, 6; 10]]_4$	$(4, 1, 3)$	EA	10	2	6	10	10	6 / n/a / > 2
$[[16, 4, 5; 8]]_4$	$(4, 1, 4)$	EA	10	4	5	8	12	5 / n/a / > 2
$[[24, 4, 9; 20]]_4$	$(4, 1, 6)$	EA	20	4	9	20	24	9 / n/a / > 2
$[[8, 4, 2; 4]]_4$	$(4, 2, 1)$	EA	4	4	2	4	6	2 / n/a / 2
$[[16, 4, 4; 12]]_4$	$(4, 2, 2)$	EA	12	4	4	12	18	4 / n/a / > 2
$[[24, 4, 6; 20]]_4$	$(4, 2, 3)$	EA	20	4	6	20	30	6 / n/a / > 2
$[[12, 6, 2; 6]]_4$	$(4, 3, 1)$	EA	6	6	2	6	10	2 / n/a / 2
$[[16, 6, 2; 10]]_4$	$(4, 4, 1)$	EA	10	6	2	10	18	2 / n/a / 2
$[[4, 2, 2; 2]]_8$	$(8, 1, 1)$	EA	2	2	2	2	2	2 / n/a / 2
$[[8, 4, 3; 4]]_8$	$(8, 1, 2)$	EA	4	4	3	4	4	3 / n/a / 3
$[[12, 2, 6; 10]]_8$	$(8, 1, 3)$	EA	10	2	6	10	10	6 / n/a / > 2
$[[12, 4, 2; 8]]_8$	$(8, 3, 1)$	EA	8	4	2	8	14	2 / n/a / 2
$[[4, 2, 2; 2]]_9$	$(9, 1, 1)$	EA	2	2	2	2	2	2 / n/a / 2
$[[8, 2, 4; 6]]_9$	$(9, 1, 2)$	EA	6	2	4	6	6	4 / n/a / > 3
$[[12, 2, 5; 6]]_9$	$(9, 1, 3)$	EA	8	2	5	6	8	5 / n/a / > 2
$[[16, 2, 8; 14]]_9$	$(9, 1, 4)$	EA	14	2	8	14	14	8 / n/a / > 2
$[[20, 4, 9; 16]]_9$	$(9, 1, 5)$	EA	16	4	9	16	16	9 / n/a / > 2
$[[8, 4, 2; 4]]_9$	$(9, 2, 1)$	EA	4	4	2	4	6	2 / n/a / 2
$[[16, 4, 4; 12]]_9$	$(9, 2, 2)$	EA	12	4	4	12	18	4 / n/a / > 2
$[[12, 4, 2; 8]]_9$	$(9, 3, 1)$	EA	8	4	2	8	14	2 / n/a / 2
$[[24, 6, 4; 18]]_9$	$(9, 3, 2)$	EA	18	6	4	18	30	– / 4 / > 2

A.2 The non-principal witnesses

The witnesses to Theorem 5.4 and the quantum codes they produce, from Part 5 of the same log:

witness	codes produced
$(9,2,3)$, $s=0$	R1: $[[24,14,2]]_9$ (abelian); R3: $[[24,6,2;2]]_9$
$(9,2,3)$, $s=1$	R1: $[[24,14,2]]_9$ (abelian); R3: $[[24,6,2;2]]_9$
$(4,2,4)$, $s=0$	R3: $[[32,8,2;4]]_4$
$(4,2,4)$, $s=1$	R3: $[[32,8,2;4]]_4$

In all four cases $\mu_B(W)=3$ and $\mu_B(\mathrm{Hull}_s(W))=3$. The two $(9,2,3)$ witnesses additionally have $\mathrm{Hull}_s(W)=W$, i.e. the code equals its own Galois hull.

A.3 Best known distances used in Section 9

Table 6 is the raw data retrieved from [5] on 2026-11-04, in the form lower..upper bound for $[[N,K]]_q$. The entry "not available" is the site's own response for $q=8$.

Table 6: Best known stabilizer distances, codetables.de, retrieved 2026-11-04. The site prints the *prime* p in its code labels, so $q=4$ appears as $[[N,K]]_2$ and $q=9$ as $[[N,K]]_3$.

$[[N,K]]_q$	lower..upper	$[[N,K]]_q$	lower..upper
$[[16,4]]_4$	5..5	$[[8,4]]_9$	3..3
$[[16,12]]_4$	2..2	$[[16,8]]_9$	4..4
$[[20,4]]_4$	6..6	$[[24,8]]_9$	6..8
$[[20,12]]_4$	3..3	$[[24,10]]_9$	6..7
$[[20,16]]_4$	2..2	$[[24,12]]_9$	5..6
$[[24,16]]_4$	3..3	$[[8,4]]_8$	not available
$[[24,20]]_4$	2..2	$[[12,2]]_8$	not available

B Reproducibility

Every number in this paper that is not proved in the text is produced by a script in the accompanying supplementary package, and every such number is attributable to a line of a committed log. This appendix records what the package contains, how to run it, and which numbers come from where.

B.1 Package layout

```
PAPER/supplementary/
  README.md                   index of the package
  ENVIRONMENT.md              exact environment used
  MANIFEST.sha256             SHA-256 of every file in the package
  scripts/                    the computation scripts (copies of the
                              committed 04_COMPUTATION/ scripts)
  logs/                       the committed logs the numbers are read from
  generator_matrices/         generator matrices of the headline codes,
                              one .json per code
  parameter_tables/           machine-readable versions of the tables of
                              this paper (CSV)
  CERTIFICATION_PROCEDURE.md what "certified" means here, and how to
                              re-certify a code from scratch
```

The scripts are byte-identical copies of the scripts committed in `04_COMPUTATION/` of the project repository; the logs are the committed logs. Nothing was re-run to produce the package, so no script hash is invalidated.

B.2 Environment

The computation is pure Python 3 standard library. No third-party package (numpy, sympy, GAP, Magma) is used or required; arithmetic in $\mathbb{F}_q$ and in A_t, R, R_n is implemented in `matrix_ring_lib.py` and is elementary by design, so that every step can be inspected. See `ENVIRONMENT.md` for the interpreter version and platform of the recorded run.

B.3 Randomness

The search uses pseudo-random sampling for the families that are too large to enumerate. The seed is fixed (`SEED=20261210`) and every random choice is drawn from a single `random.Random(SEED)` instance, in a fixed order of families, so a re-run reproduces every line of the log except the per-family elapsed-time annotations and the peak-memory line, which are wall-clock and resource readings and take no part in any computation.

The principal results of this paper do not depend on the sampling: the theorems are proved, and the headline codes come from families that are enumerated exhaustively.

B.4 Where the numbers come from

- Theorems 4.3–7.9, 8.1, 8.3, 8.5, 8.7, 6.3, 6.13, 6.10, 7.3, 5.4: proved in the text; no computation enters.
- Table 1 (the $\mu_B = 3$ witnesses) and the four witness codes of Appendix A: Part 5 of `phase7_extended_search_log.txt`, produced by `phase7_extended_search.py`.
- Table 5 (all 28 certified codes): Part 6 of the same log.
- Table 2: Part 4(d) (Pareto frontier), Part 4(e) (best per route) and Part 6 of the same log.
- Table 3 (minimum entanglement per distance): Part 4(c) of the same log.
- Table 4 and Table 6: Part 8 of the same log for the $q = 4$ rows, and the pages of [5] retrieved on 2026-11-04 for the $q = 9$ rows and for the bounds themselves.
- The counts 120 721, 5 055, 4 418, 637, 368, 88, 280, 25 families scanned, 42 families in the census (23 exhaustive, 4 counted-only, 15 blocked): the completion report and Part 2 of the same log.
- The bookkeeping identities of Theorem 8.7 are proved, and check A1 of the log verifies them on all 5 055 codes (0 violations). The coset Singleton bound is check H1 (0 violations). The Singleton and Hamming bounds are check G1 (0 violations).

B.5 How to reproduce

```
cd PAPER/supplementary/scripts
python3 phase7_extended_search.py > out.txt 2>&1
diff <(grep -v '\[[0-9]*s\]' out.txt) \
     <(grep -v '\[[0-9]*s\]' ../logs/phase7_extended_search_log.txt)
```

The run takes about 9 minutes and about 370 MB of resident memory on the recorded platform. The `grep -v` filter removes the elapsed-time annotations; the peak-memory line is the only other difference.

The earlier phases are reproduced the same way from their own scripts; see `04_COMPUTATION/README.md` in the repository for the dependency order (`matrix_ring_lib.py` first, then the phase scripts).

B.6 What is *not* reproducible from the package

The $q = 9$ bounds of Table 6 were read from a web page that changes; the retrieved values are dated 2026-11-04 and are quoted verbatim, and the page reported "Last change: 10.06.2024". They are external data, not an output of this computation.

B.7 Verification of the package itself

```
cd PAPER/supplementary
sha256sum -c MANIFEST.sha256
```

Every file listed in the manifest is checked against its recorded hash; see `PAPER/SHA256_MANIFEST.md` for the same check over the manuscript and the whole package.

References

[1] Adel Alahmadi, Houda Sboui, Patrick Solé, and Olfa Yemen. Cyclic codes over $M_2(\mathbb{F}_2)$. *Journal of the Franklin Institute*, 350(9):2837–2847, 2013.

[2] Meng Cao. MDS codes with Galois hulls of arbitrary dimensions and the related entanglement-assisted quantum error correction. *IEEE Transactions on Information Theory*, 67(12):7964–7984, 2021.

[3] Hai Q. Dinh, Atul Gaur, Pratyush Kumar, Manoj Kumar Singh, and Abhay Kumar Singh. Cyclic codes over rings of matrices. *Advances in Mathematics of Communications*, 18(4):1100–1122, 2024.

[4] Yun Fan and Liang Zhang. Galois self-dual constacyclic codes. *Designs, Codes and Cryptography*, 84(3):473–492, 2017.

[5] Markus Grassl. Bounds on the minimum distance of linear codes and quantum codes. Online table, http://www.codetables.de, 2024.

[6] Marcus Greferath and Sergio R. Schmidt. Gray isometries for finite chain rings and a nonlinear ternary $(36, 3^{12}, 15)$ code. *IEEE Transactions on Information Theory*, 45(7):2522–2524, 1999.

[7] Soham Ravikant Joshi, Shikha Patel, and Om Prakash. Construction of Cyclic Codes over a Class of Matrix Rings. *arXiv e-print 2602.18255 [cs.IT]*, 2026.

[8] Andreas Ketkar, Andreas Klappenecker, Santosh Kumar, and Pradeep Kiran Sarvepalli. Nonbinary stabilizer codes over finite fields. *IEEE Transactions on Information Theory*, 52(11):4892–4914, 2006.

[9] Hongwei Liu and Xu Pan. Galois hulls of linear codes over finite fields. *Designs, Codes and Cryptography*, 88(2):241–255, 2020.

[10] Rong Luo and Udaya Parampalli. Cyclic codes over $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$. *Cryptography and Communications*, 10(6):1109–1117, 2018.

[11] Xuesong Si and Chuanze Niu. On skew cyclic codes over $M_2(\mathbb{F}_2)$. *AIMS Mathematics*, 8(10):24434–24445, 2023.

[12] Jay A. Wood. Duality for modules over finite rings and applications to coding theory. *American Journal of Mathematics*, 121(3):555–575, 1999.

[13] Enbin Zhang, Bo Kong, and Xiying Zheng. Quantum Codes from Galois Hulls of Constacyclic Codes over a Finite Non-Chain Ring. *Entropy*, 28(4):407, 2026.